Appendix A. Additional Results: ACS Datasets

A.1 Income Task – Arguably Causal

A.1.1 CA Train Domain

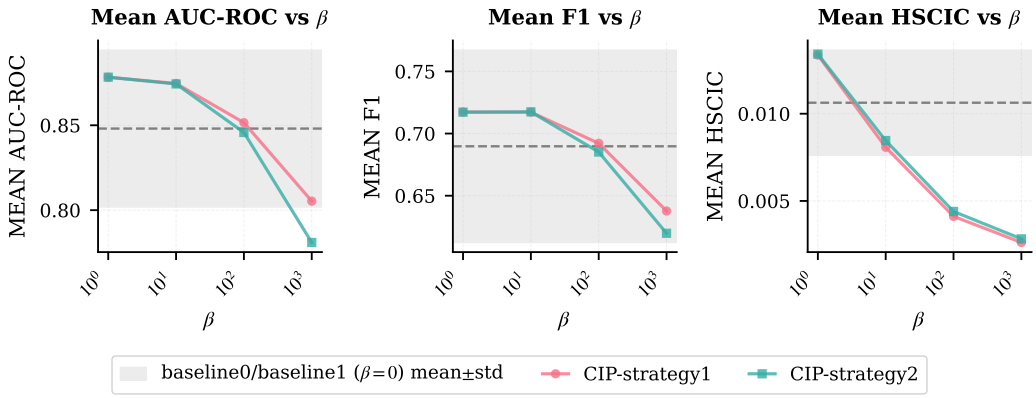


Figure A.1: Income task, arguably causal CIP features, CA training domain: mean AUC-ROC, Macro-F1 and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

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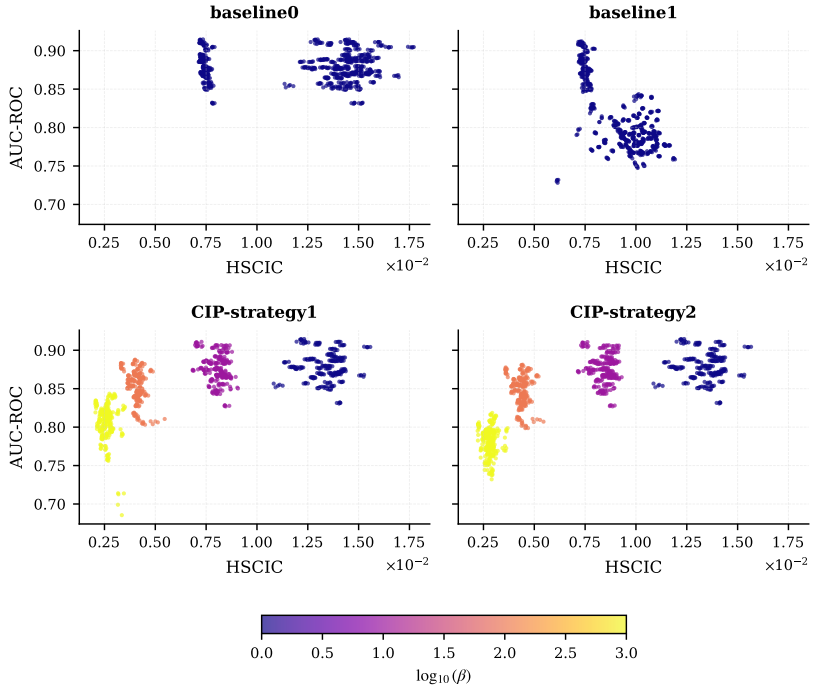


Figure A.2: Income task, arguably causal CIP features, CA training domain: relation between HSCIC and AUC-ROC across baseline and CIP strategies.

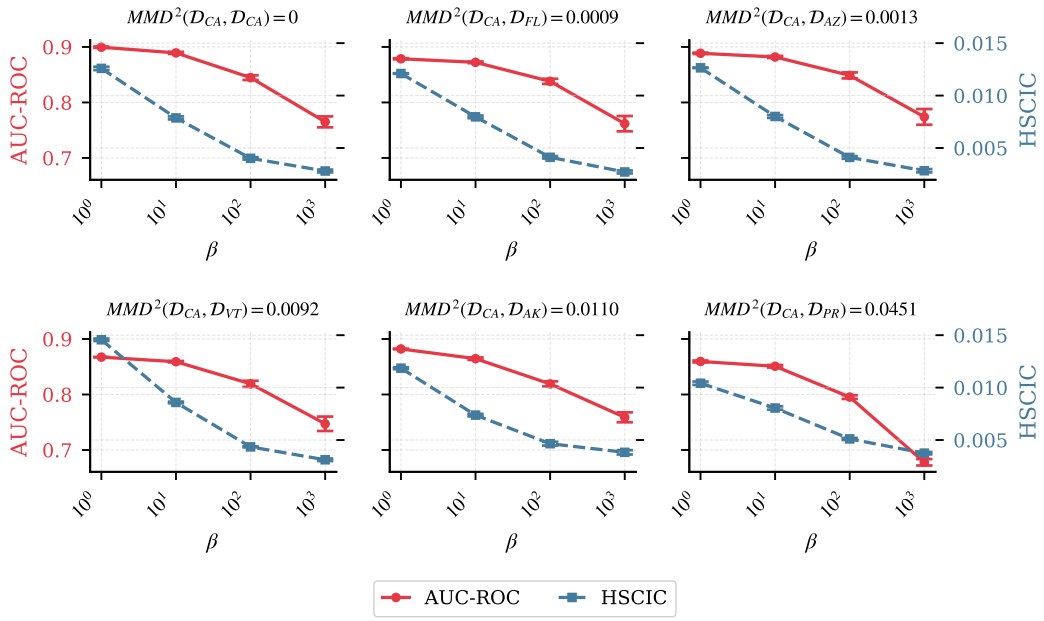


Figure A.3: Income task, arguably causal CIP features, CA training domain: relation between HSCIC and AUC-ROC versus β for Strategy 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

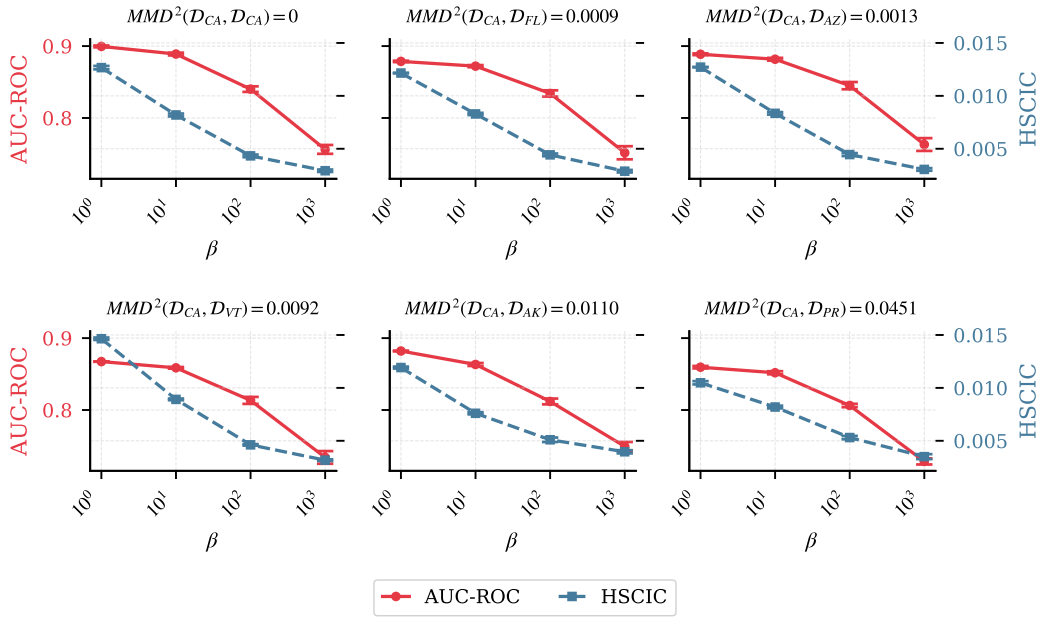


Figure A.4: Income task, arguably causal CIP features, CA training domain: relation between HSCIC and AUC-ROC versus β for Strategy 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

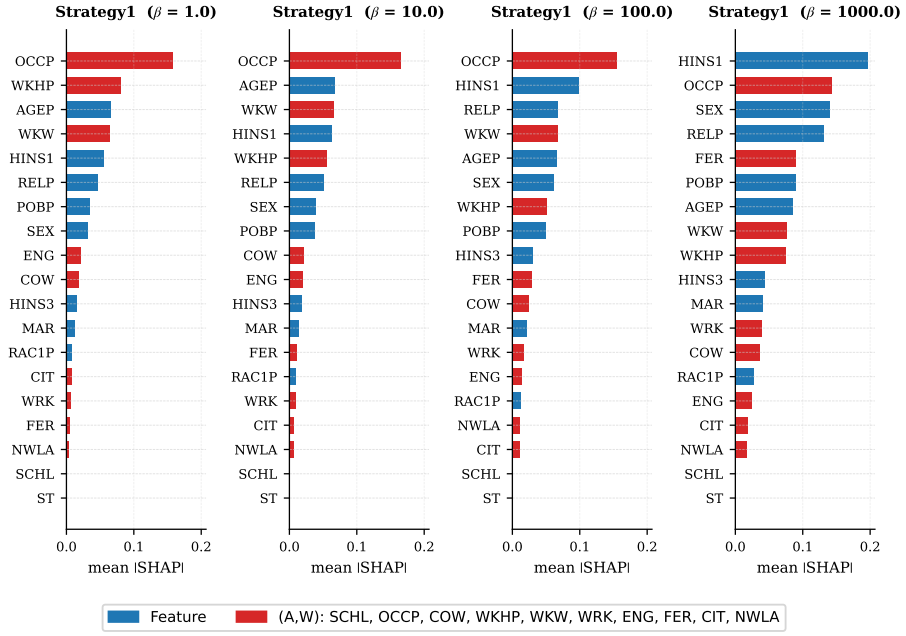


Figure A.5: SHAP analysis: income task, arguably causal CIP features, CA train domain (out-of-domain), STRATEGY 1 (joint optimisation). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

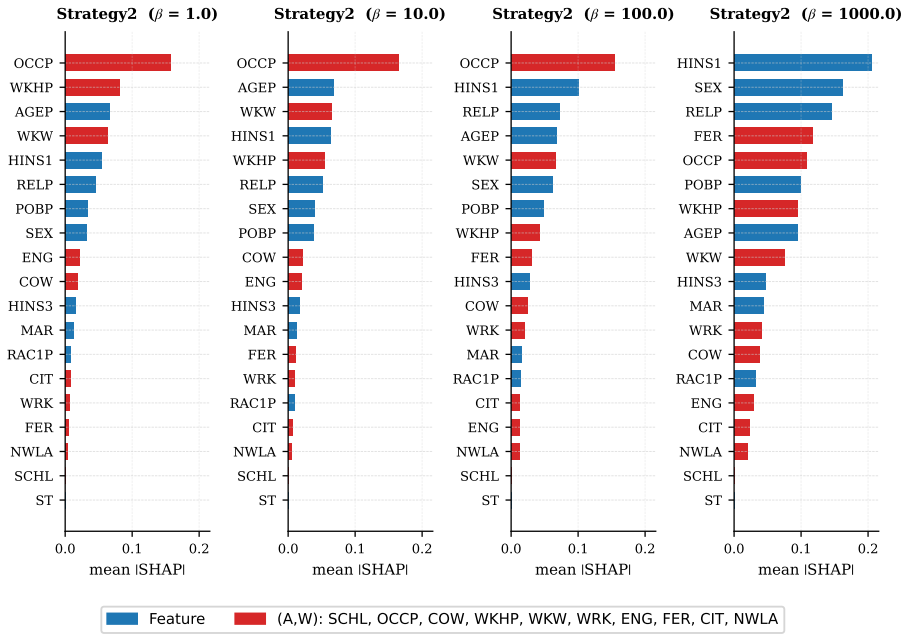


Figure A.6: SHAP analysis: income task, arguably causal CIP features, CA train domain (out-of-domain), STRATEGY 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

A.1.2 TX Train Domain

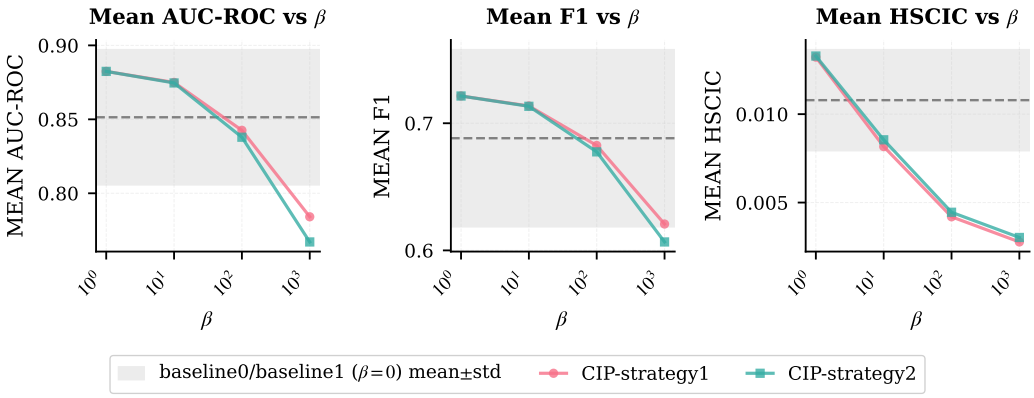


Figure A.7: Income task, arguably causal CIP features, TX training domain: mean AUC-ROC, Macro-F1 and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

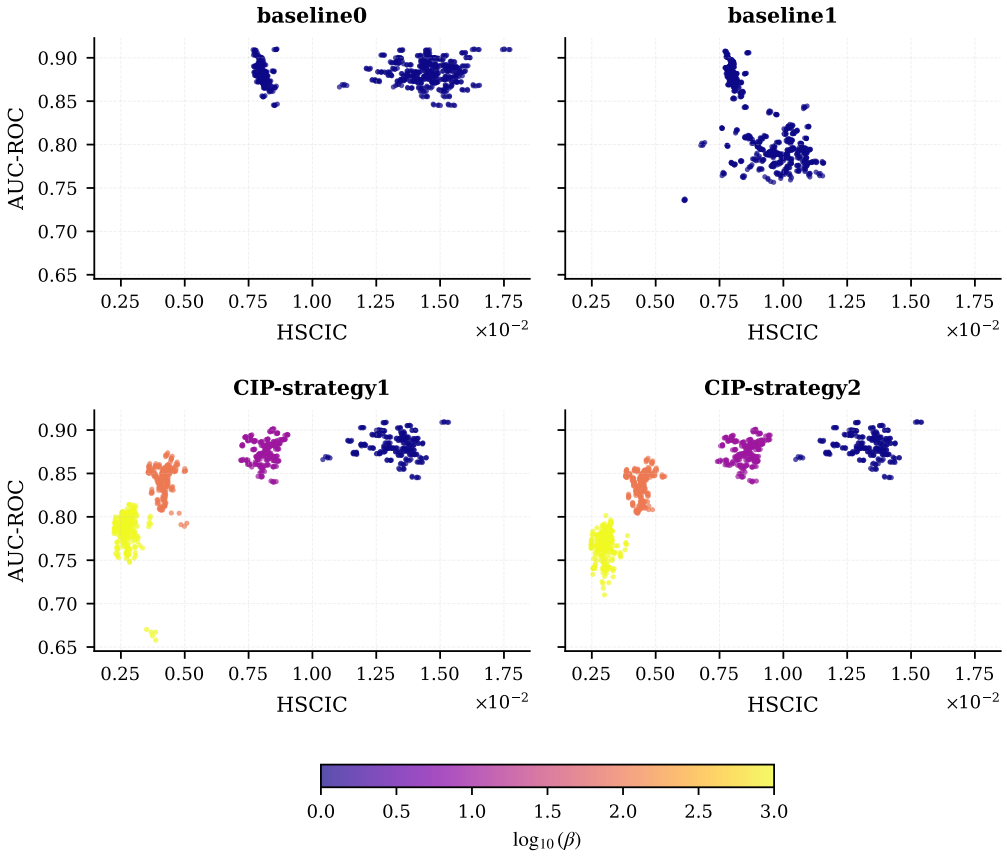


Figure A.8: Income task, arguably causal CIP features, TX training domain: relation between HSCIC and AUC-ROC across baseline and CIP strategies.

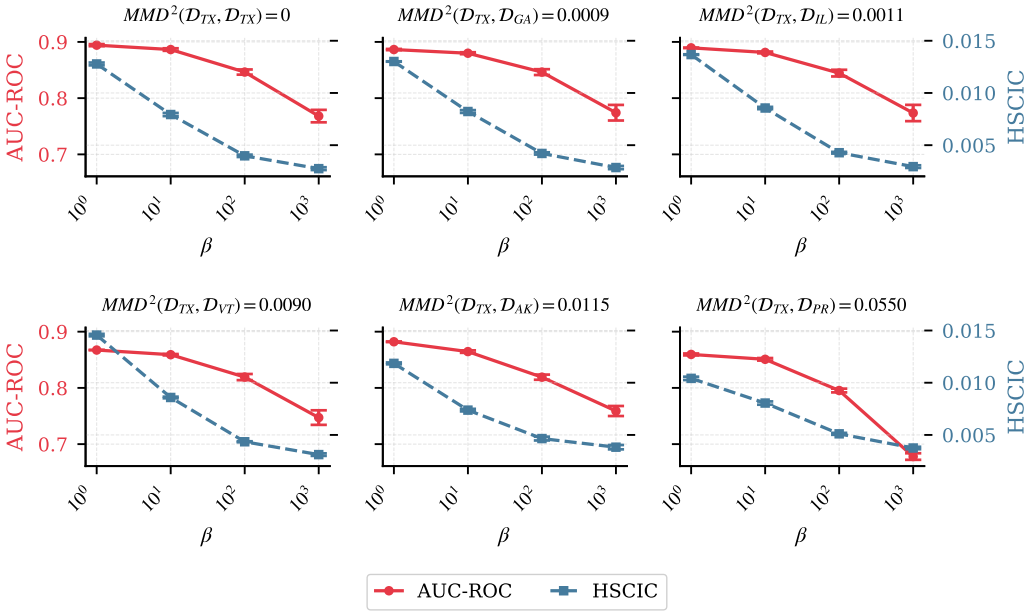


Figure A.9: Income task, arguably causal CIP features, TX training domain: relation between HSCIC and AUC-ROC versus β for Strategy 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

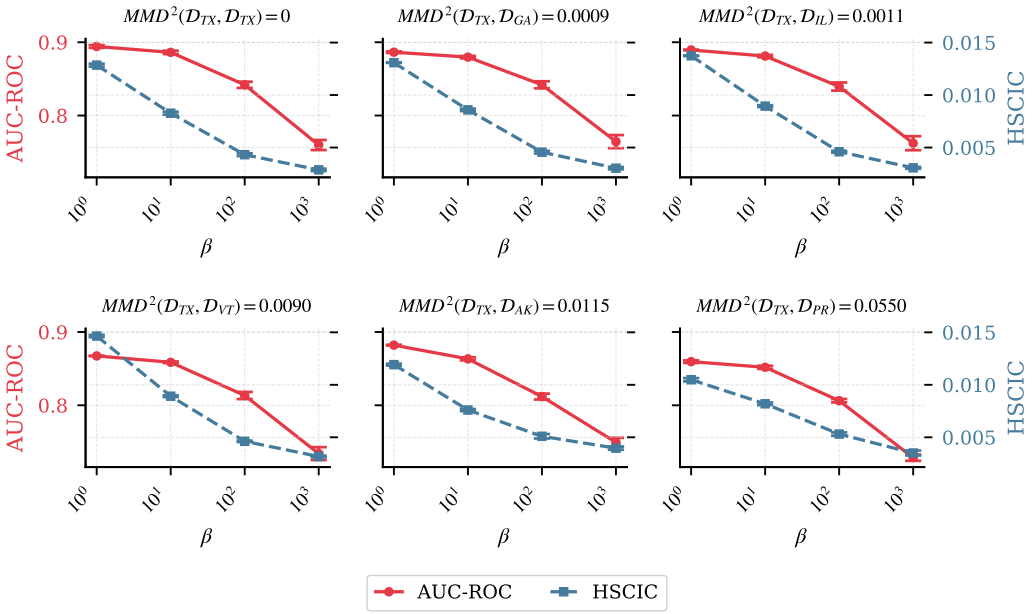


Figure A.10: Income task, arguably causal CIP features, TX training domain: relation between HSCIC and AUC-ROC versus β for Strategy 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

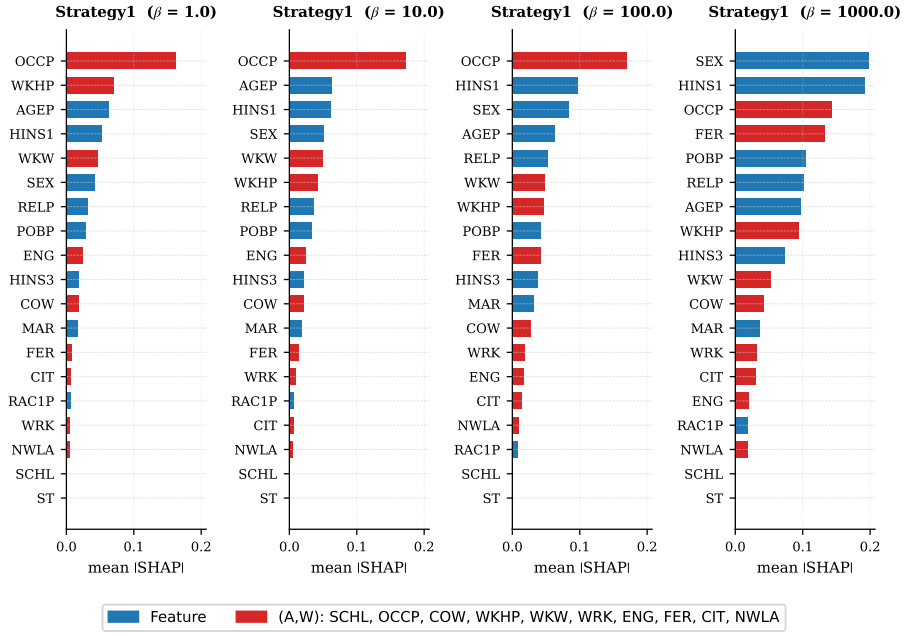


Figure A.11: SHAP analysis: income task, arguably causal CIP features, TX train domain (out-of-domain), STRATEGY 1 (joint optimisation). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

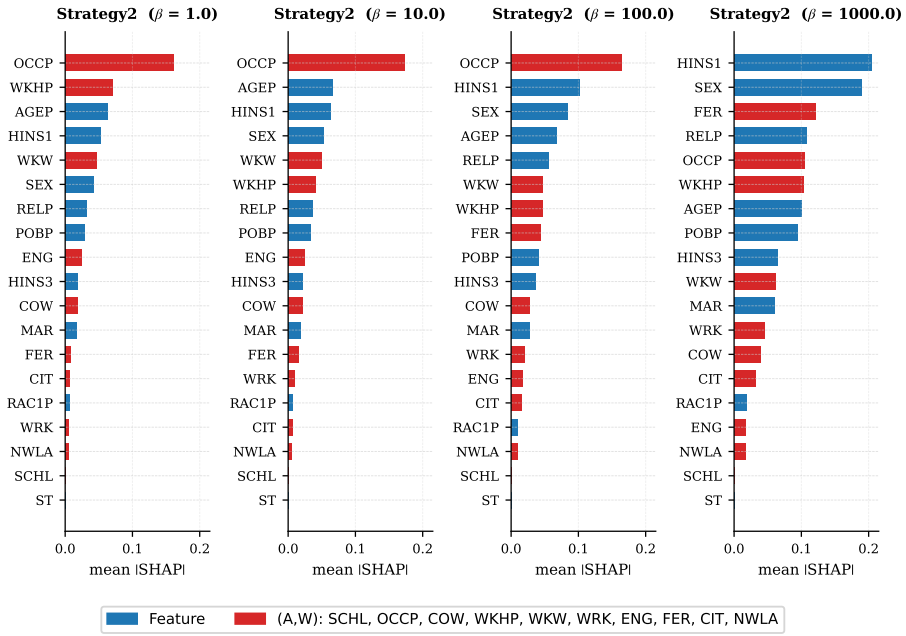


Figure A.12: SHAP analysis: income task, arguably causal CIP features, TX train domain (out-of-domain), STRATEGY 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

A.2 Income Task – State Dependent

A.2.1 CA Train Domain

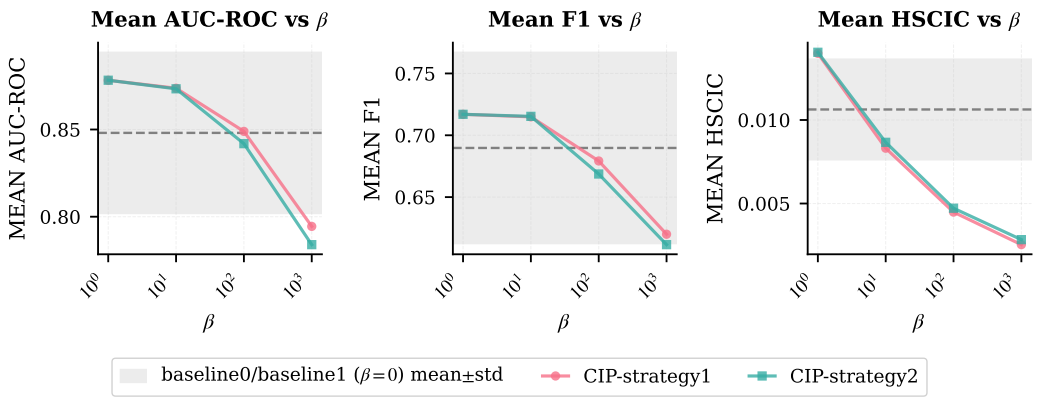


Figure A.13: Income task, state-dependent CIP features, CA training domain: mean AUC-ROC, Macro-F1 and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

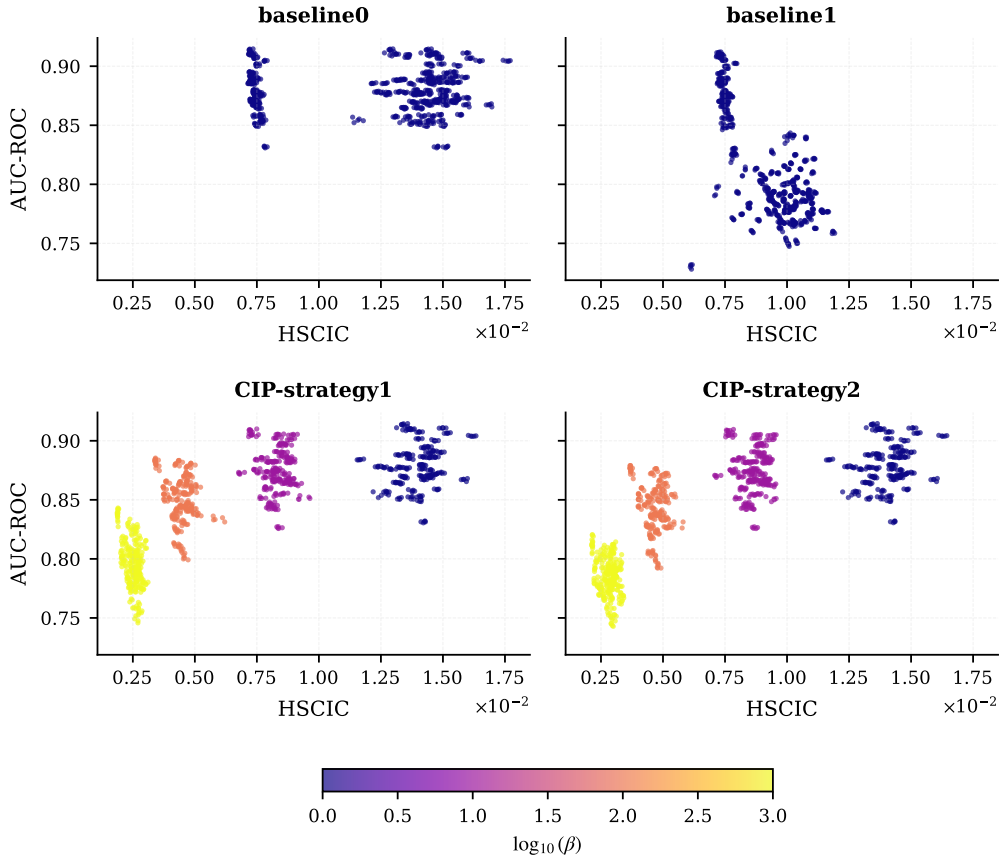


Figure A.14: Income task, state-dependent CIP features, CA training domain: relation between HSCIC and AUC-ROC across baseline and CIP strategies.

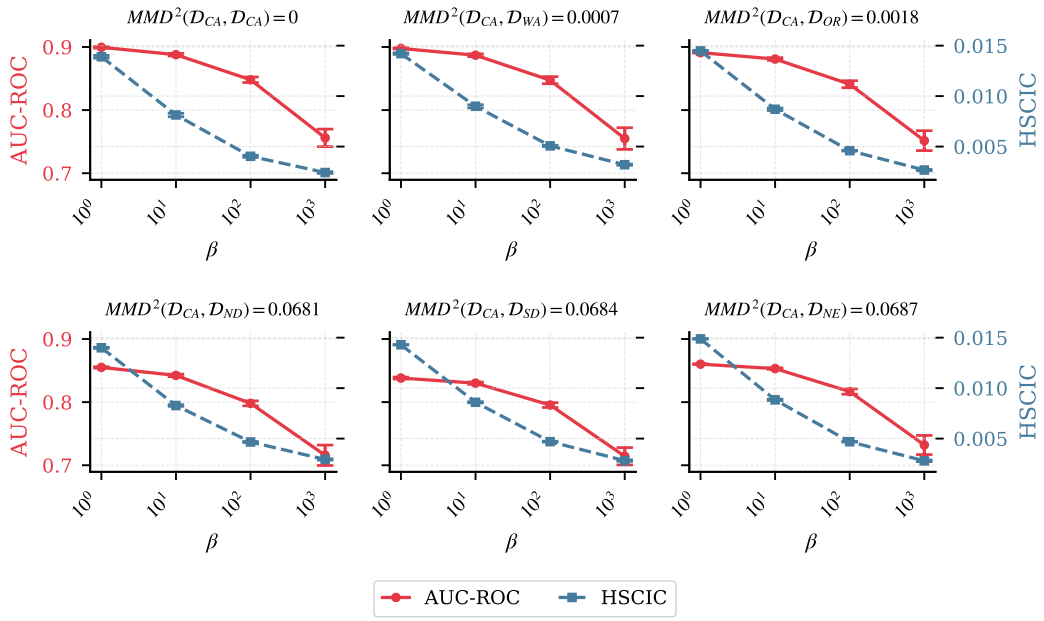


Figure A.15: Income task, state-dependent CIP features, CA training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

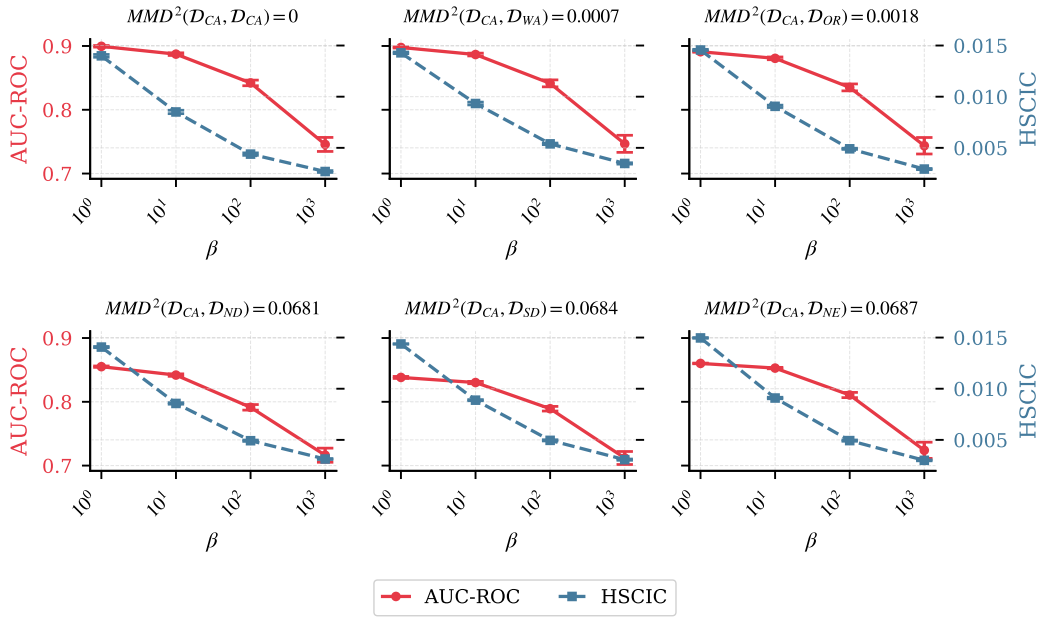


Figure A.16: Income task, state-dependent CIP features, CA training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

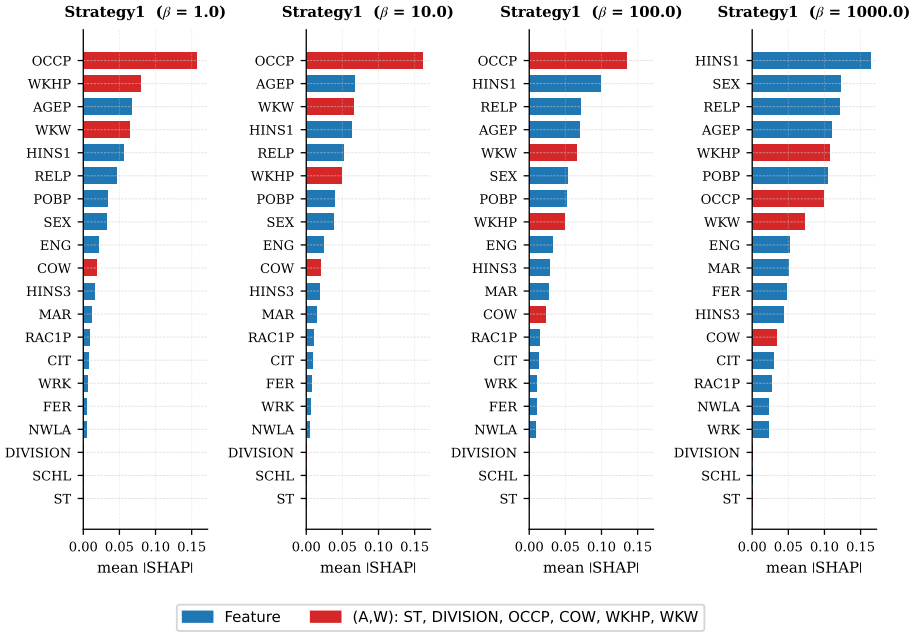


Figure A.17: SHAP analysis: income task, state-dependent CIP features, CA train domain (out-of-domain), STRATEGY 1 (joint optimisation). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

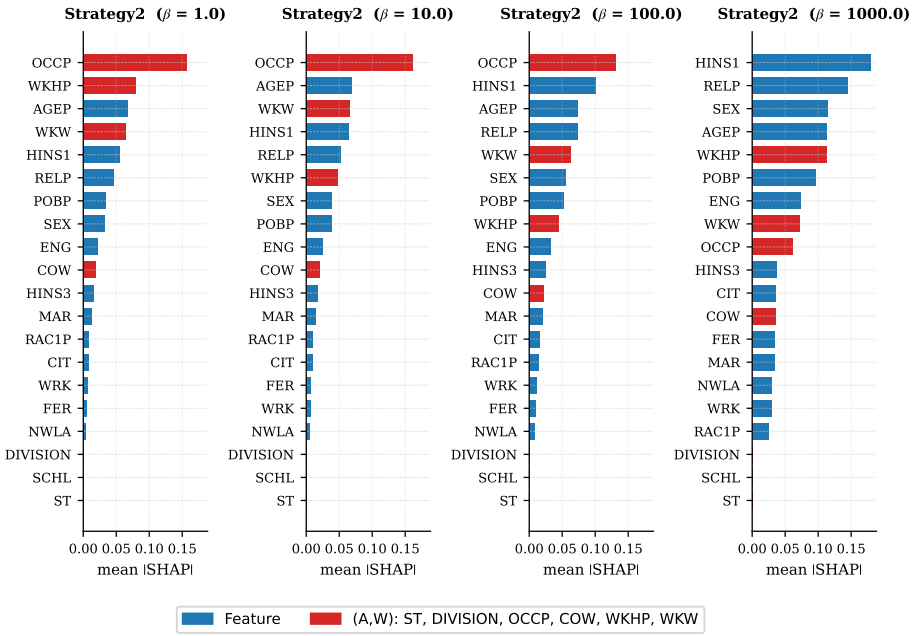


Figure A.18: SHAP analysis: income task, state-dependent CIP features, CA train domain (out-of-domain), STRATEGY 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

A.2.2 MI Train Domain

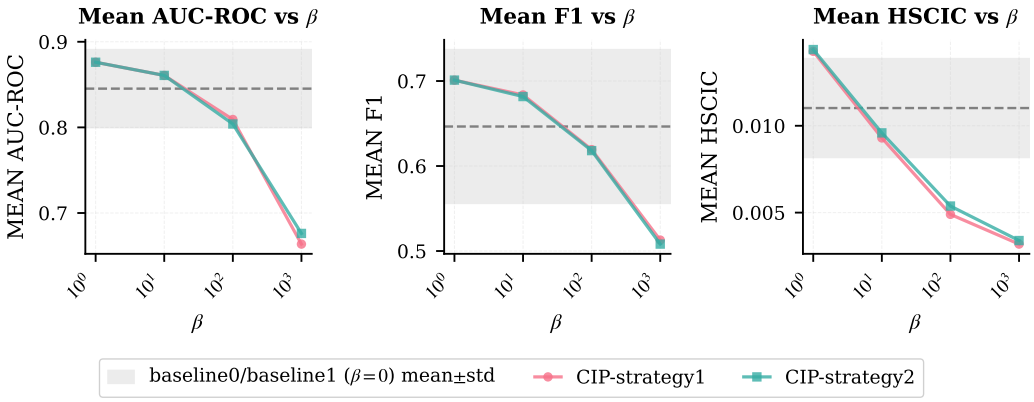


Figure A.19: Income task, state-dependent CIP features, MI training domain: mean AUC-ROC, Macro-F1 and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

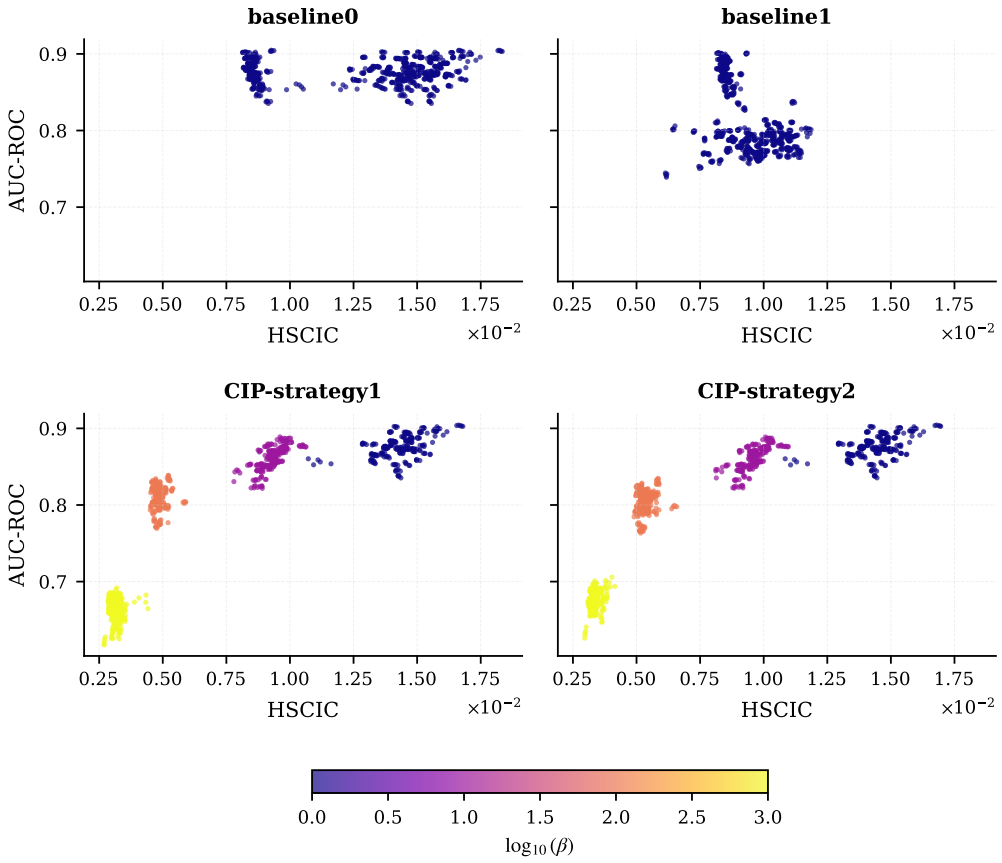


Figure A.20: Income task, state-dependent CIP features, MI training domain: relation between HSCIC and AUC-ROC across baseline and CIP strategies.

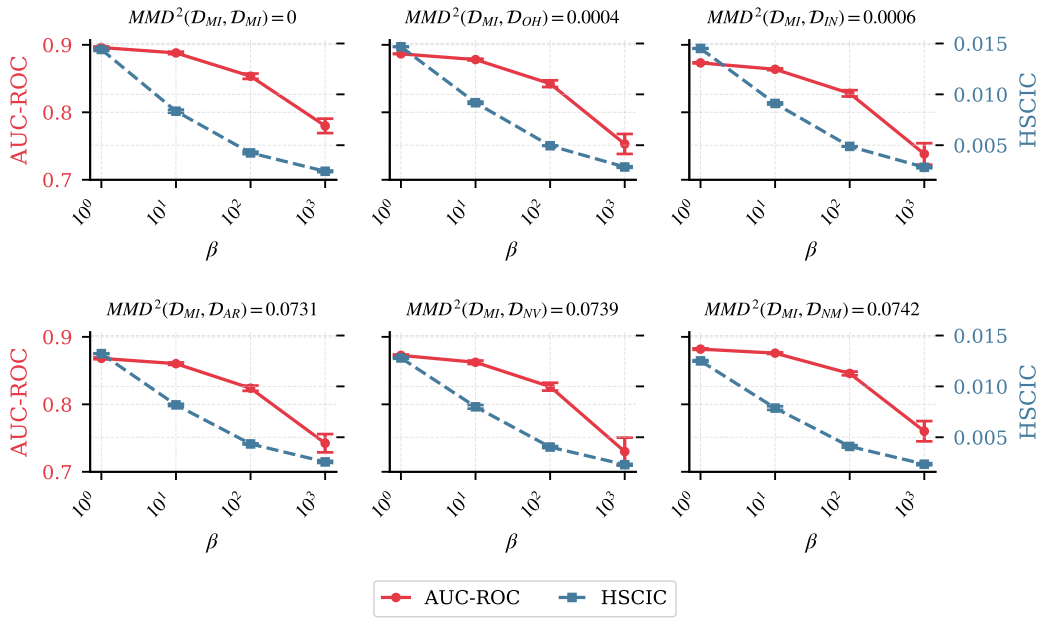


Figure A.21: Income task, state-dependent CIP features, MI training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

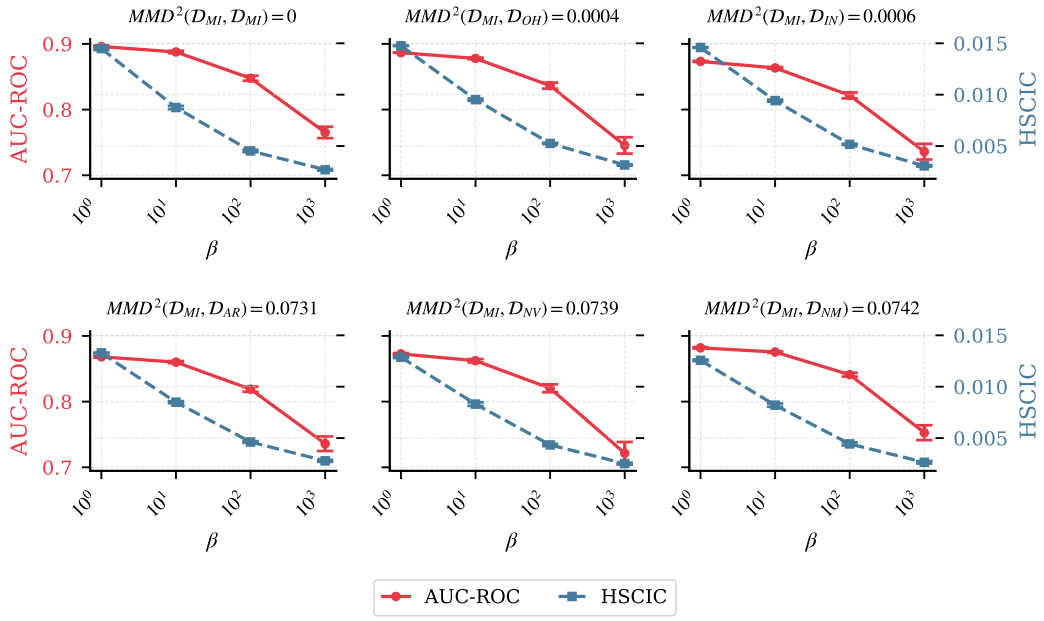


Figure A.22: Income task, state-dependent CIP features, MI training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

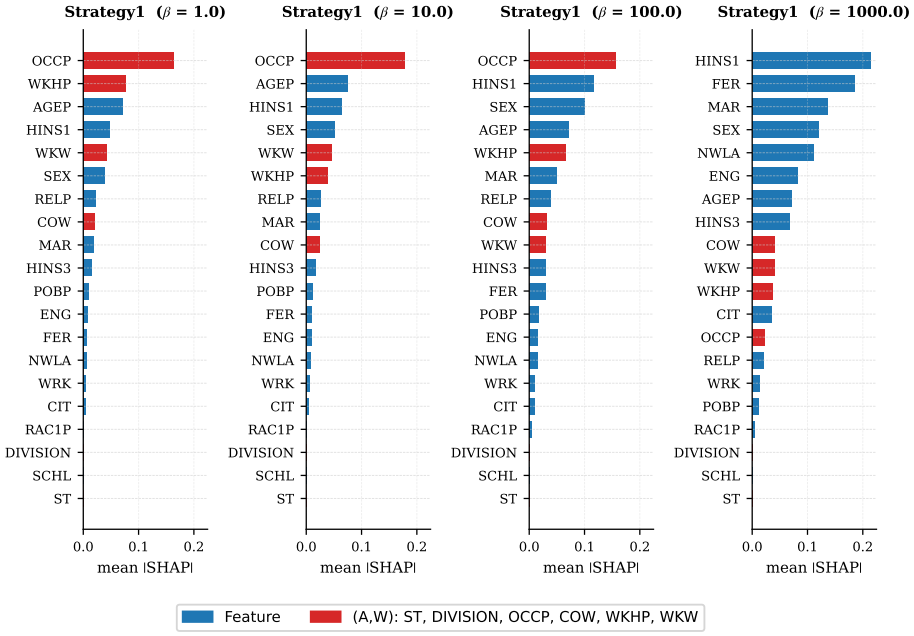


Figure A.23: SHAP analysis: income task, state-dependent CIP features, MI train domain (out-of-domain), STRATEGY 1 (joint optimisation). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

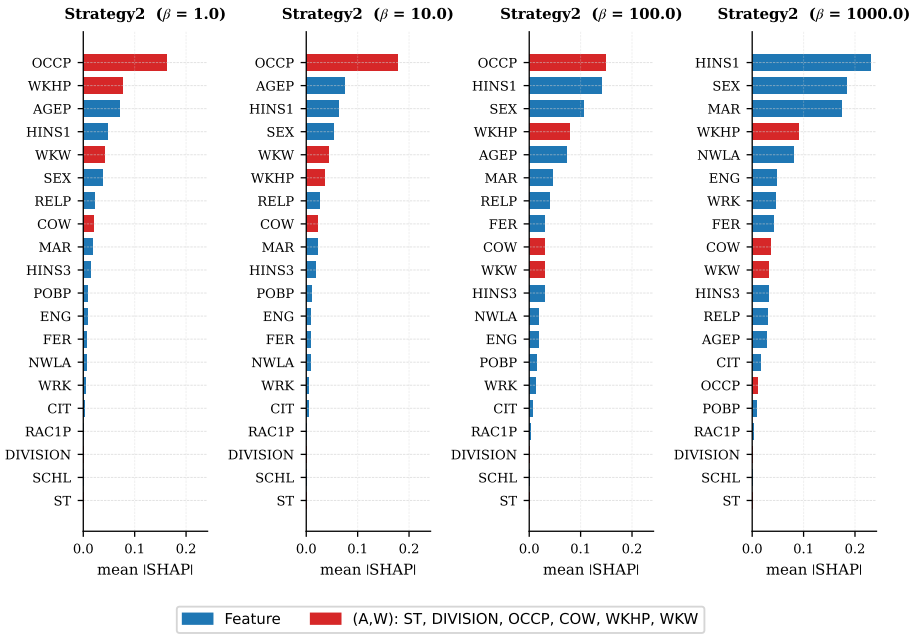


Figure A.24: SHAP analysis: income task, state-dependent CIP features, MI train domain (out-of-domain), STRATEGY 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

A.2.3 TX Train Domain

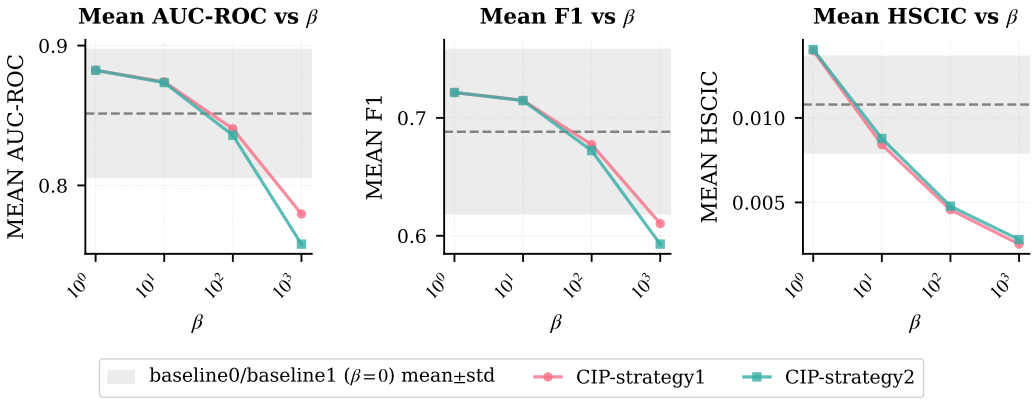


Figure A.25: Income task, state-dependent CIP features, TX training domain: mean AUC-ROC, Macro-F1 and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

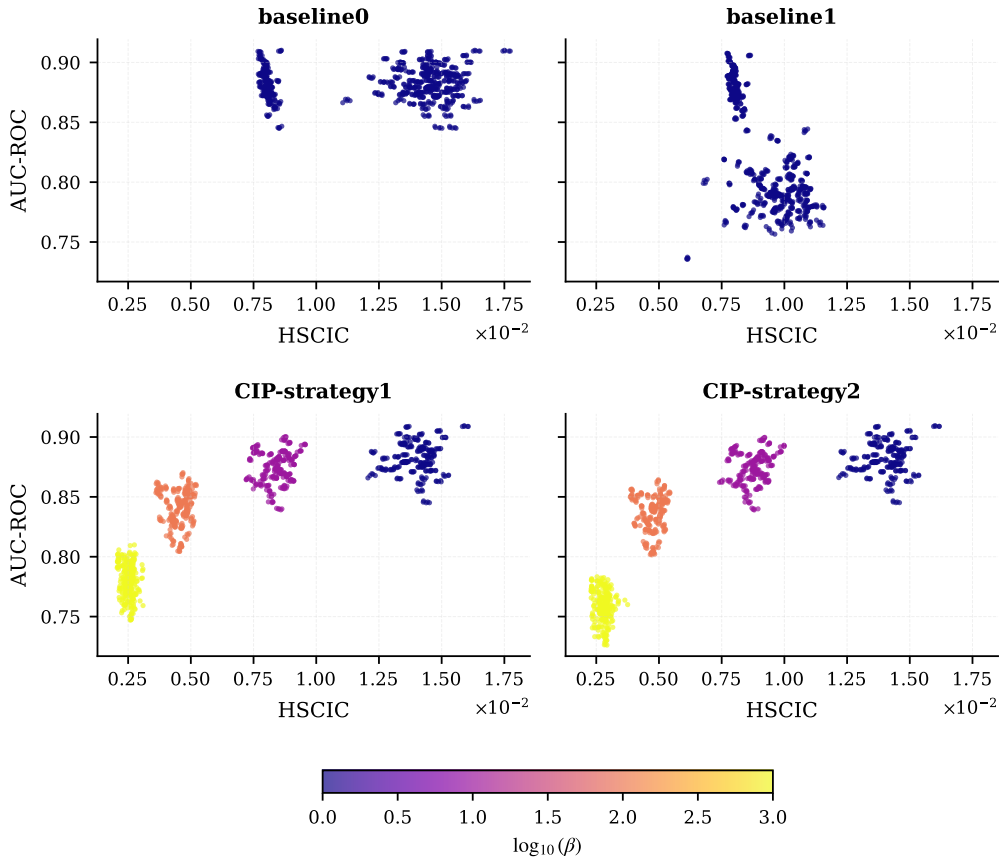


Figure A.26: Income task, state-dependent CIP features, TX training domain: relation between HSCIC and AUC-ROC across baseline and CIP strategies.

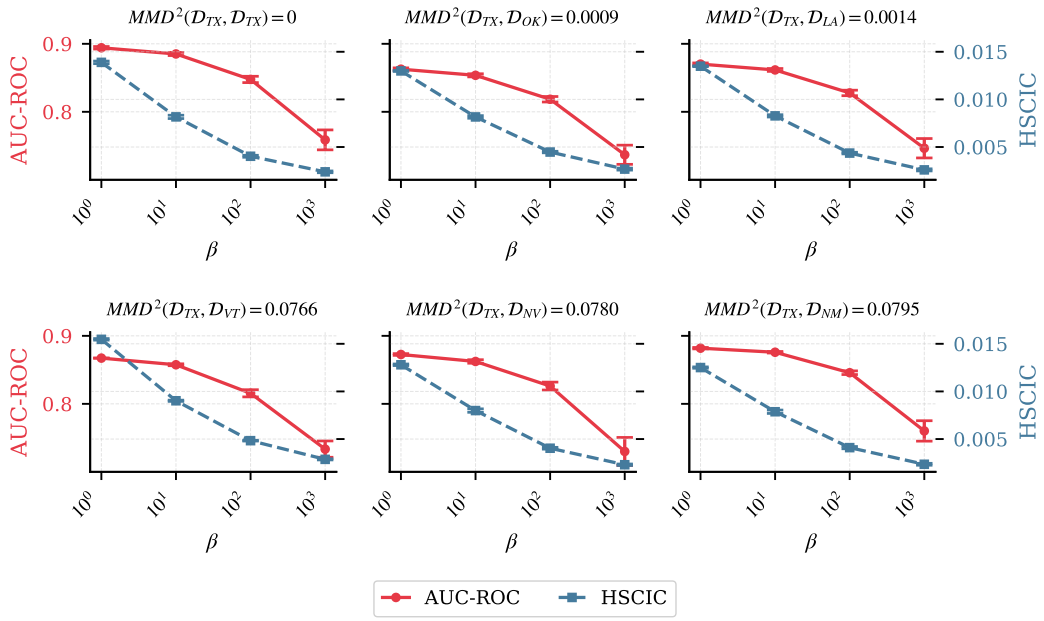


Figure A.27: Income task, state-dependent CIP features, TX training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

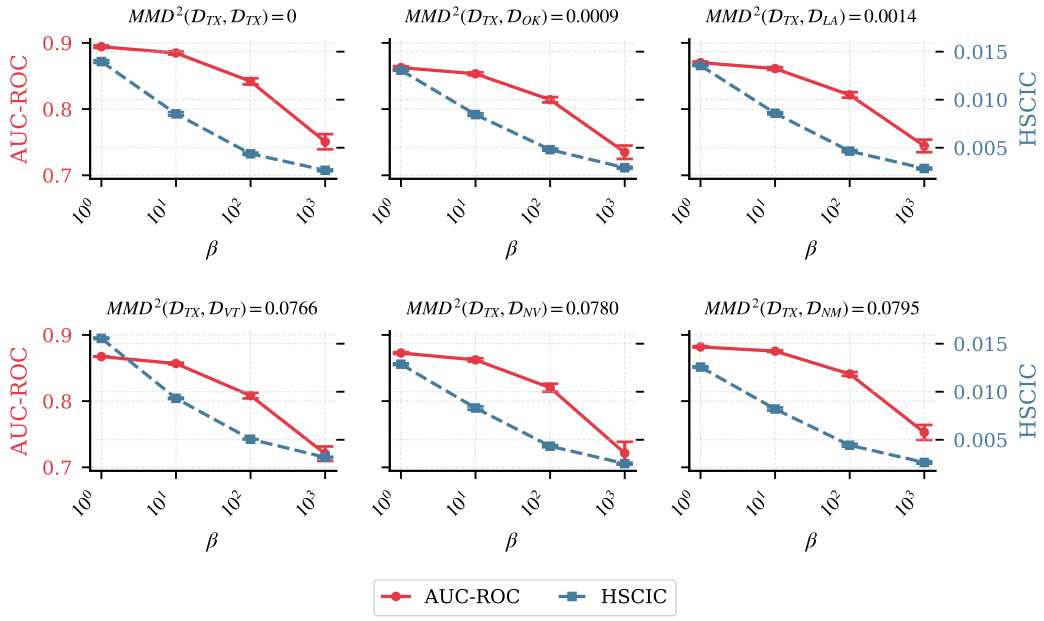


Figure A.28: Income task, state-dependent CIP features, TX training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

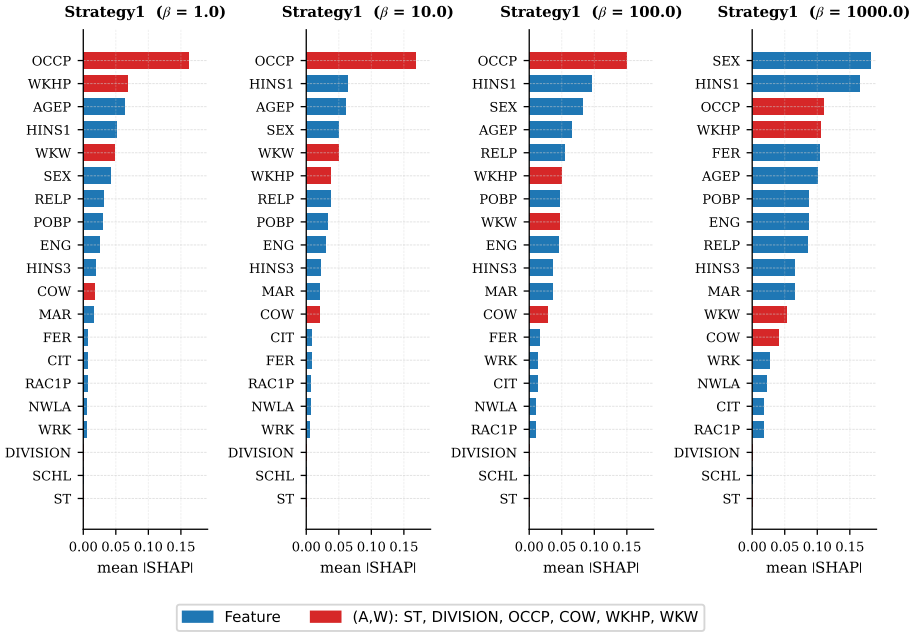


Figure A.29: SHAP analysis: income task, state-dependent CIP features, TX train domain (out-of-domain), STRATEGY 1 (joint optimisation). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

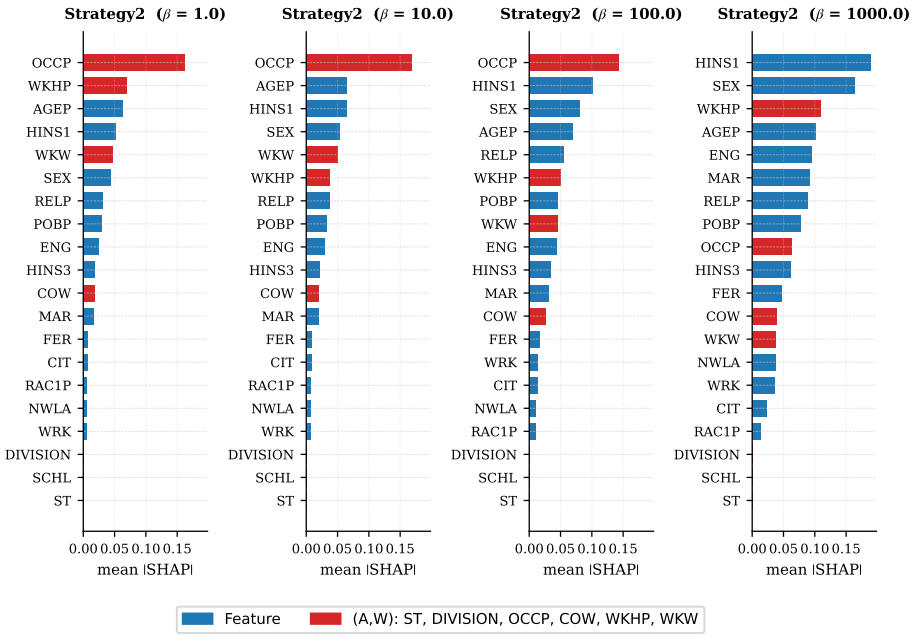


Figure A.30: SHAP analysis: income task, state-dependent CIP features, TX train domain (out-of-domain), STRATEGY 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

A.3 Income Task – Fairness

A.3.1 CA Train Domain

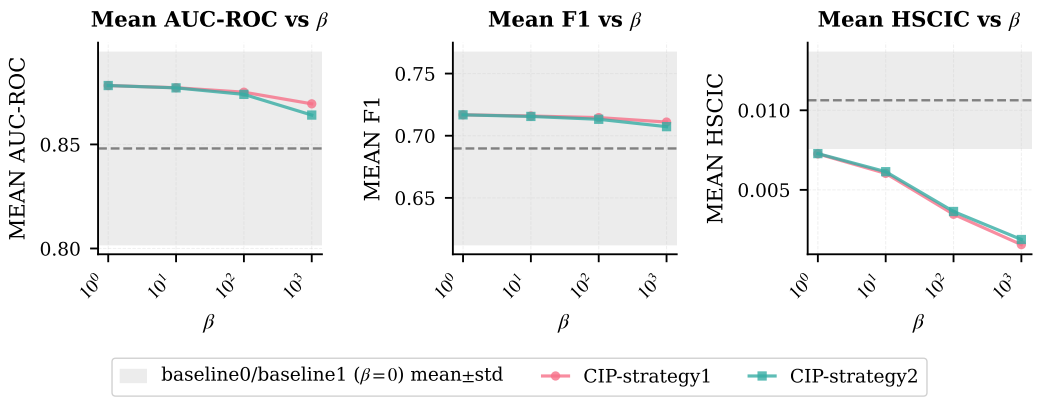


Figure A.31: Income task, fairness CIP features, CA training domain: mean AUC-ROC, Macro-F1 and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

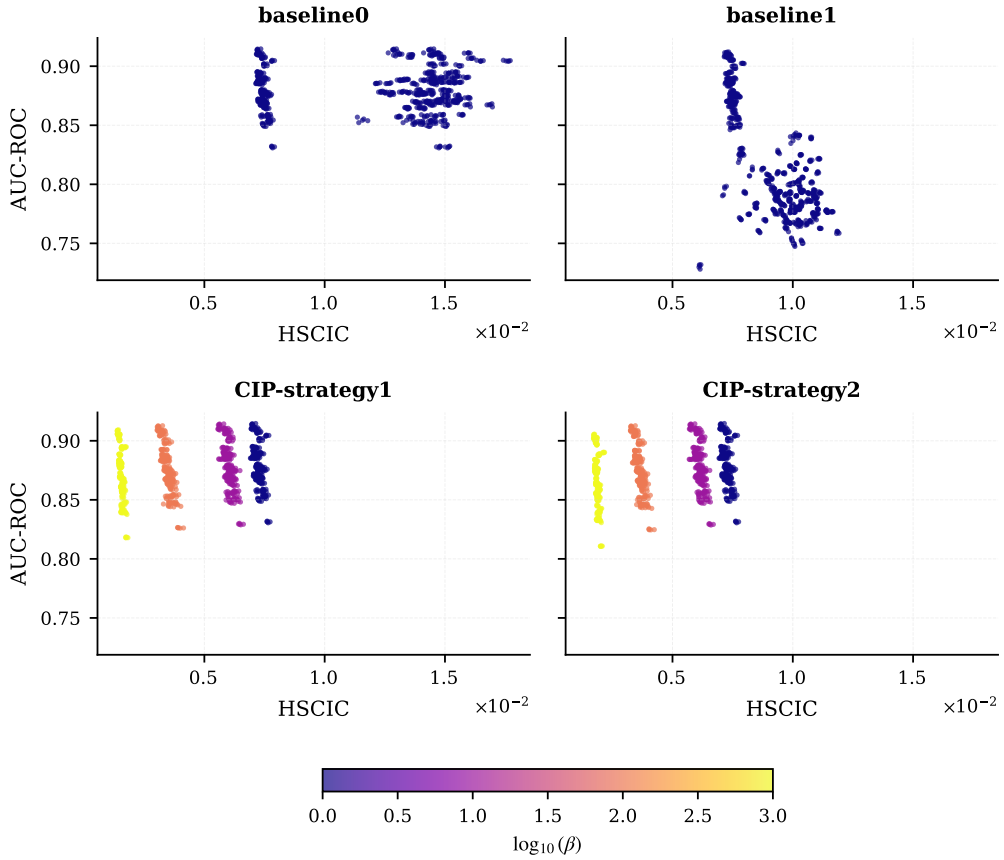


Figure A.32: Income task, fairness CIP features, CA training domain: relation between HSCIC and AUC-ROC across baseline and CIP strategies.

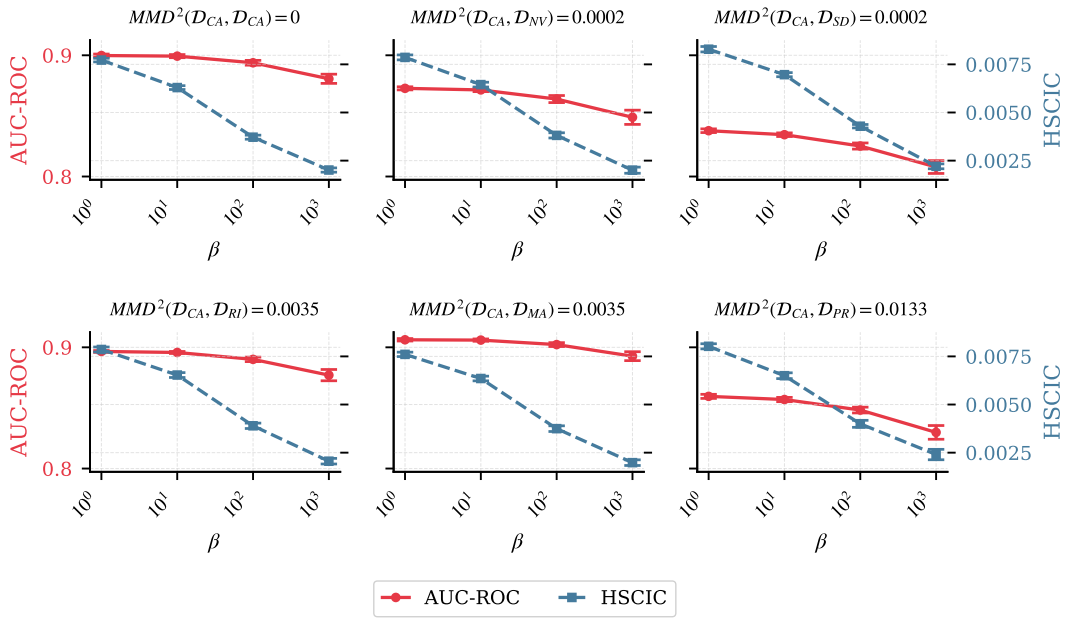


Figure A.33: Income task, fairness CIP features, CA training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

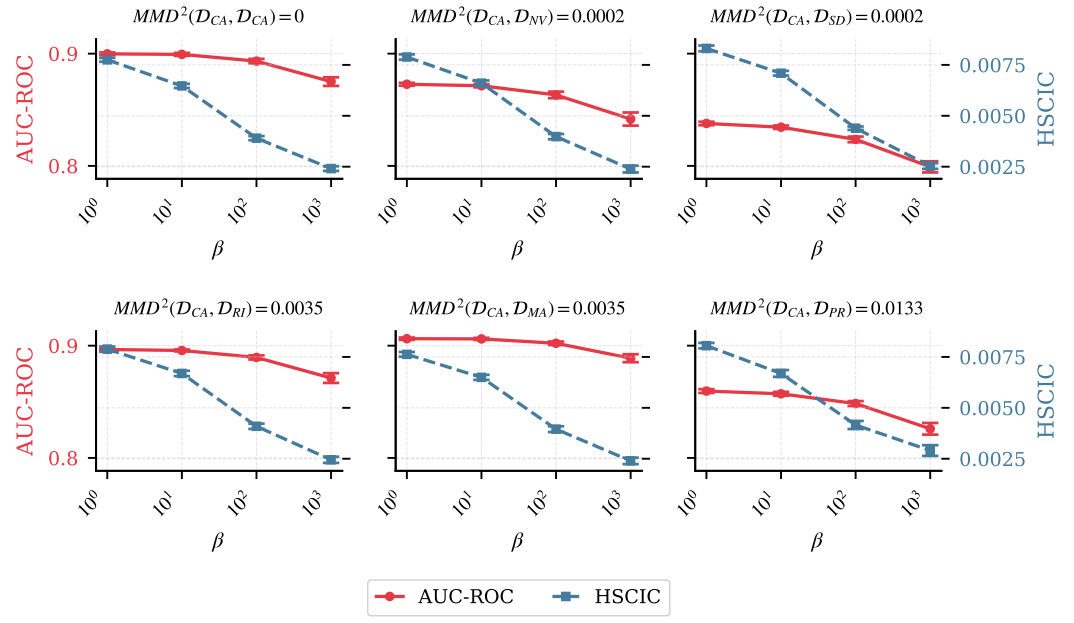


Figure A.34: Income task, fairness CIP features, CA training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

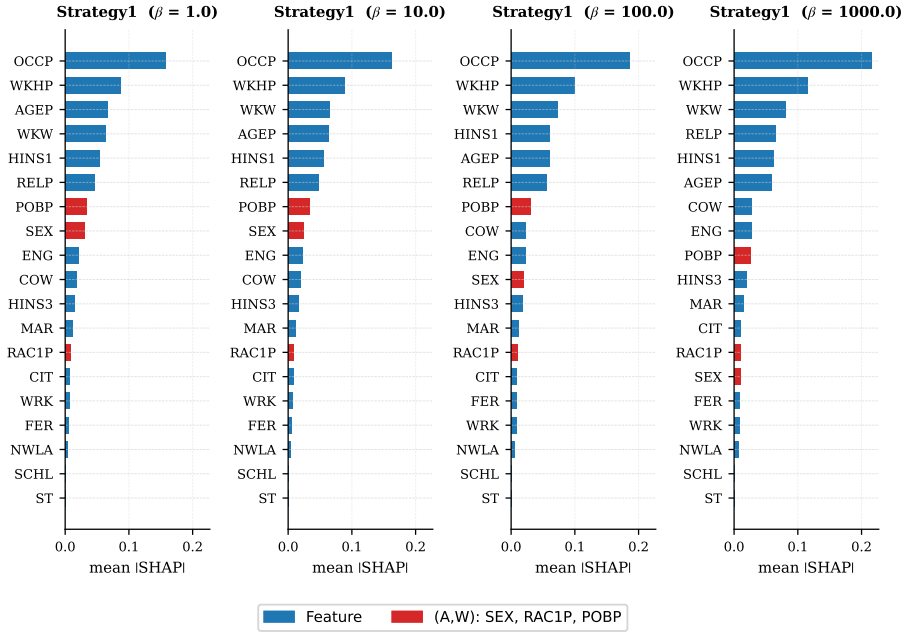


Figure A.35: SHAP analysis: income task, fairness CIP features, CA train domain (out-of-domain), STRATEGY 1 (joint optimisation). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

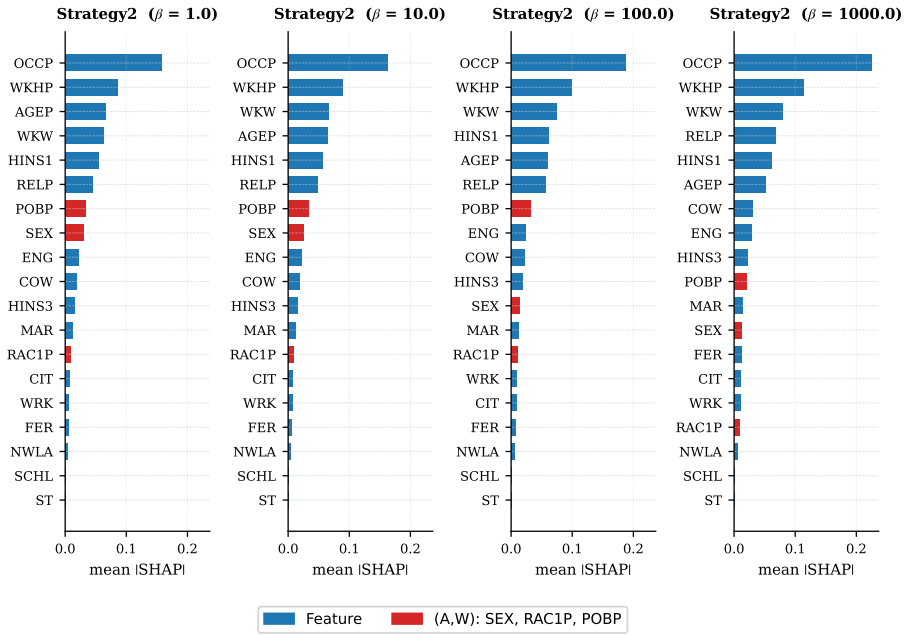


Figure A.36: SHAP analysis: income task, fairness CIP features, CA train domain (out-of-domain), STRATEGY 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

A.3.2 MI Train Domain

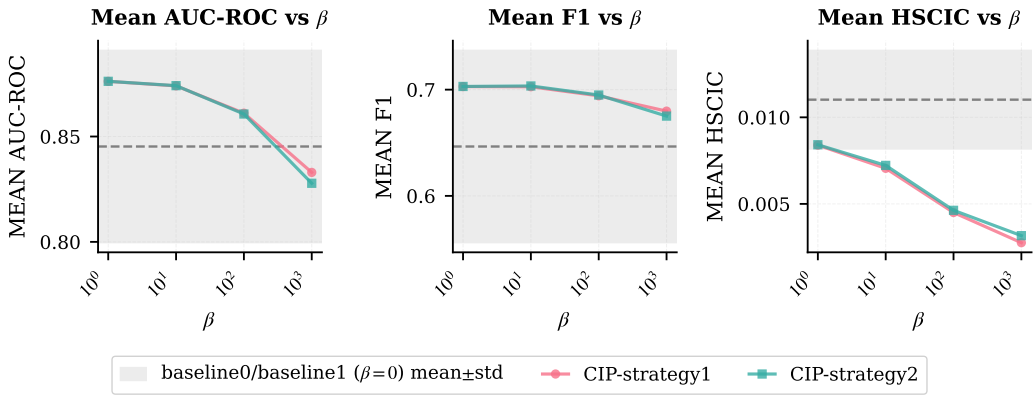


Figure A.37: Income task, fairness CIP features, MI training domain: mean AUC-ROC, Macro-F1 and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

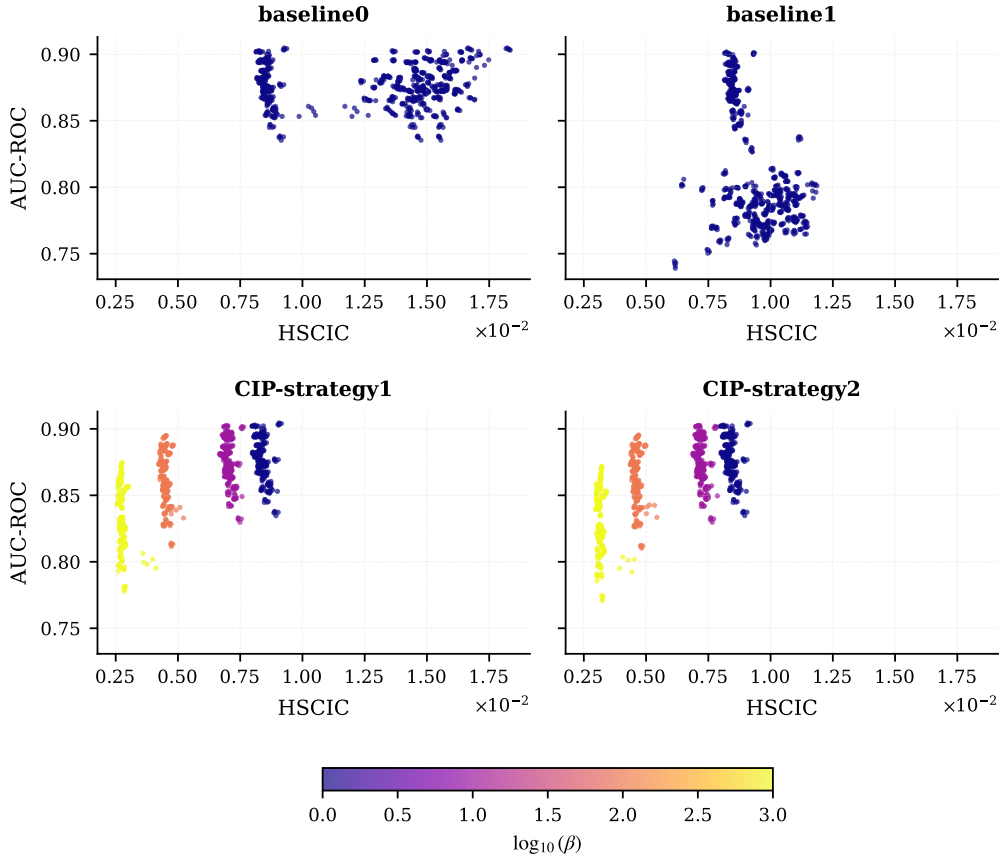


Figure A.38: Income task, fairness CIP features, MI training domain: relation between HSCIC and AUC-ROC across baseline and CIP strategies.

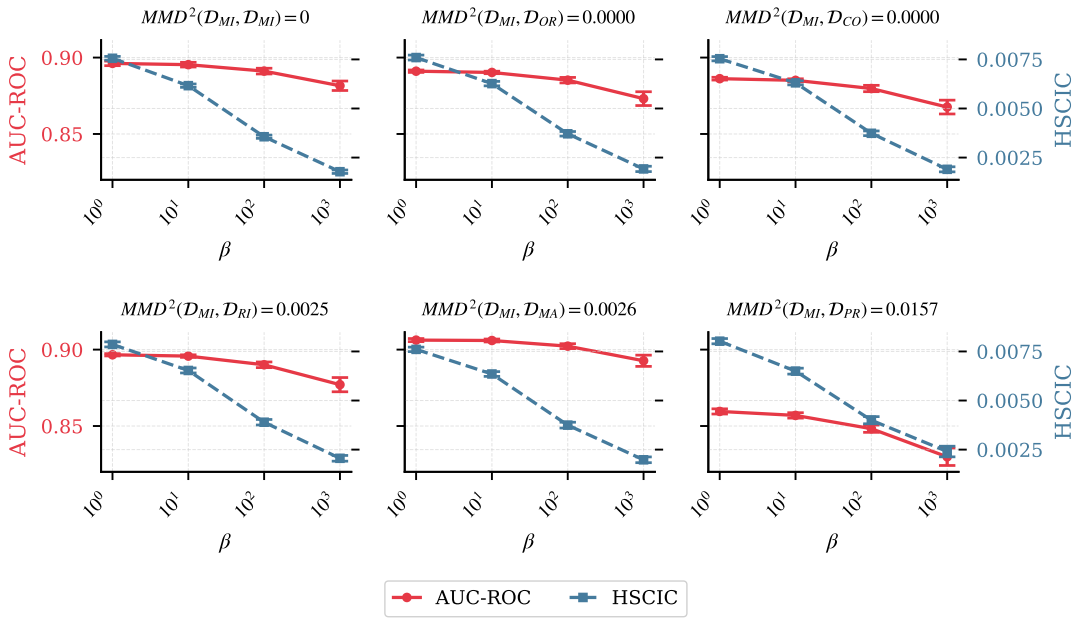


Figure A.39: Income task, fairness CIP features, MI training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

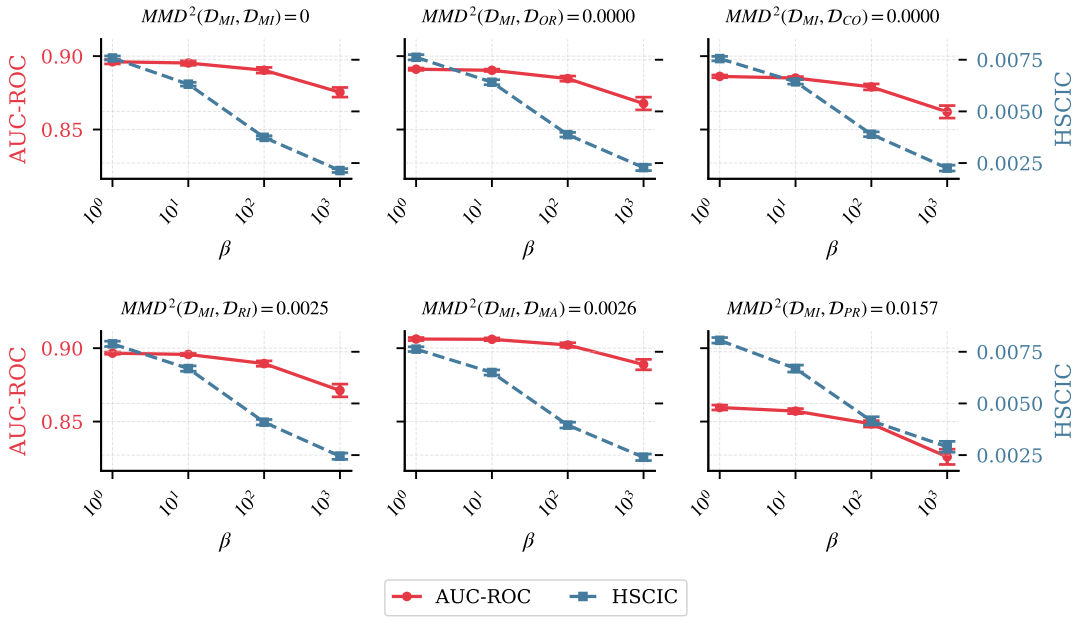


Figure A.40: Income task, fairness CIP features, MI training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

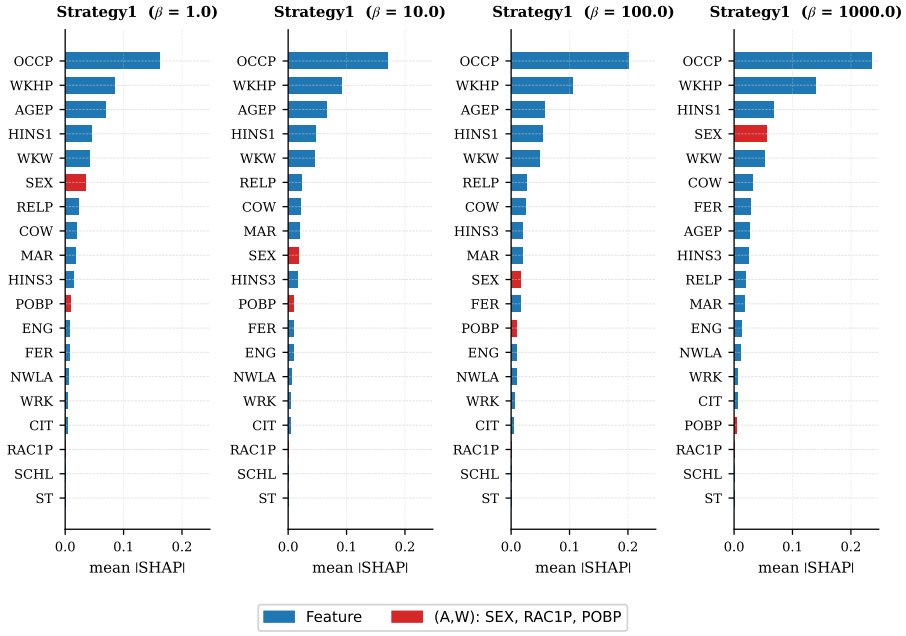


Figure A.41: SHAP analysis: income task, fairness CIP features, MI train domain (out-of-domain), STRATEGY 1 (joint optimisation). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

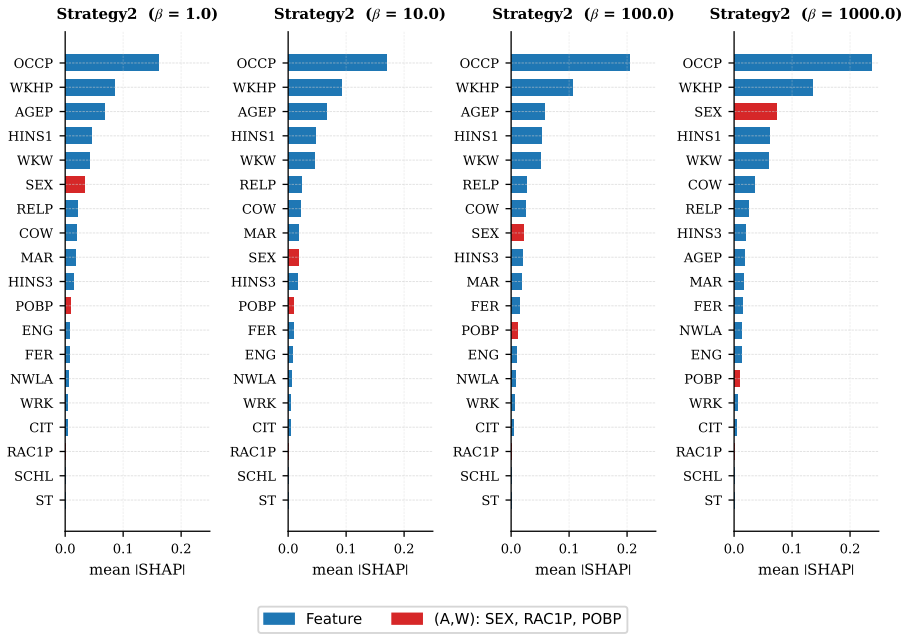


Figure A.42: SHAP analysis: income task, fairness CIP features, MI train domain (out-of-domain), STRATEGY 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

A.3.3 TX Train Domain

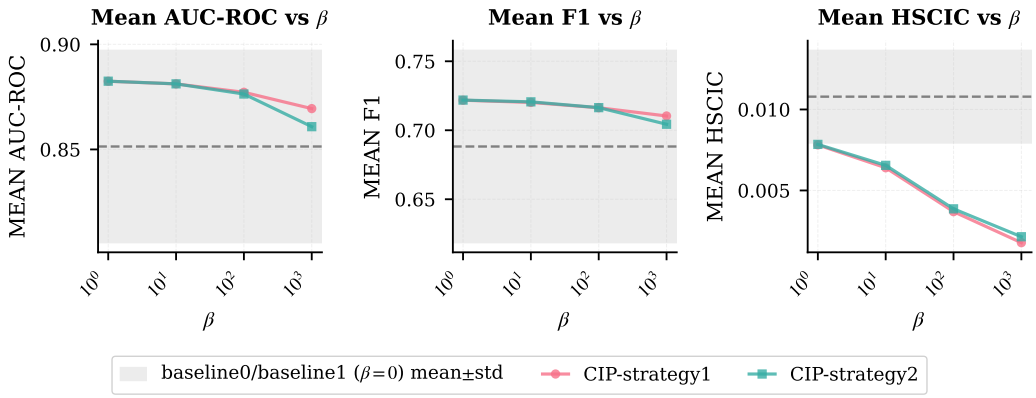


Figure A.43: Income task, fairness CIP features, TX training domain: mean AUC-ROC, Macro-F1 and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

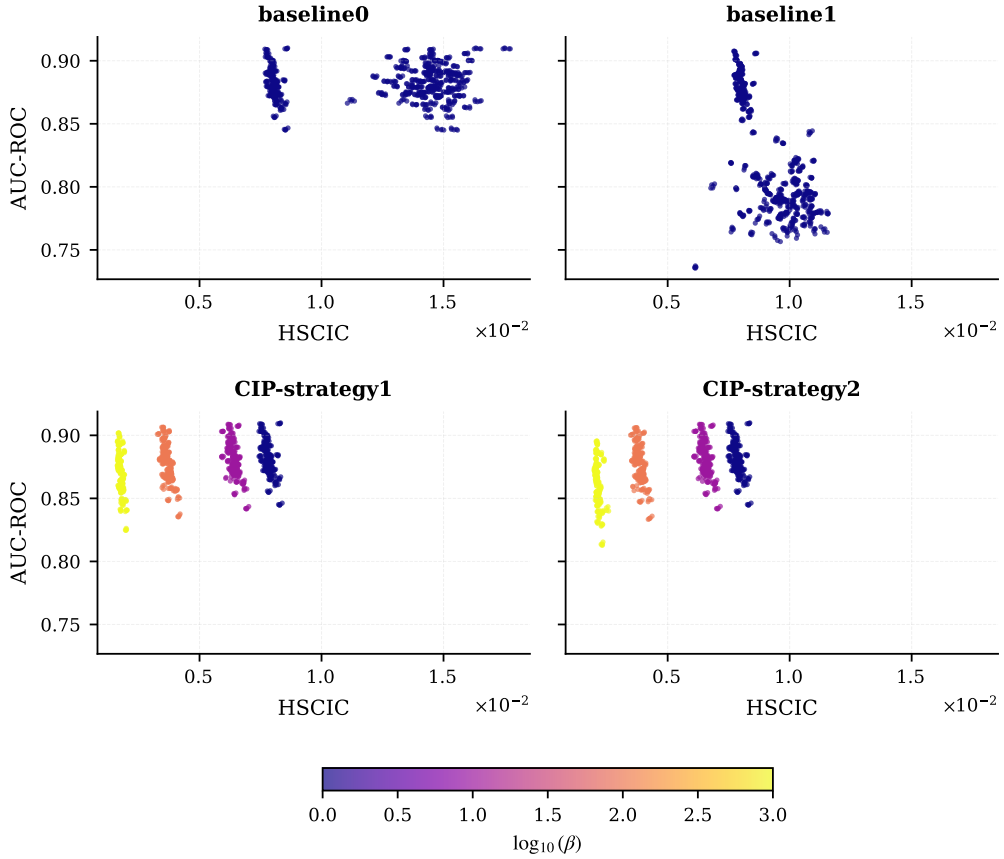


Figure A.44: Income task, fairness CIP features, TX training domain: relation between HSCIC and AUC-ROC across baseline and CIP strategies.

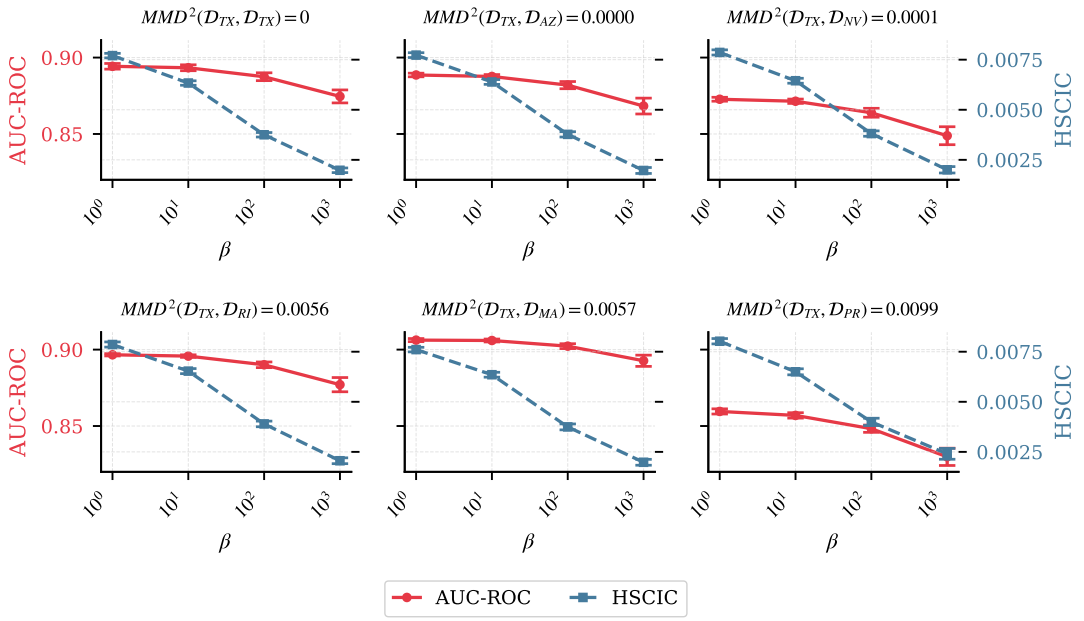


Figure A.45: Income task, fairness CIP features, TX training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

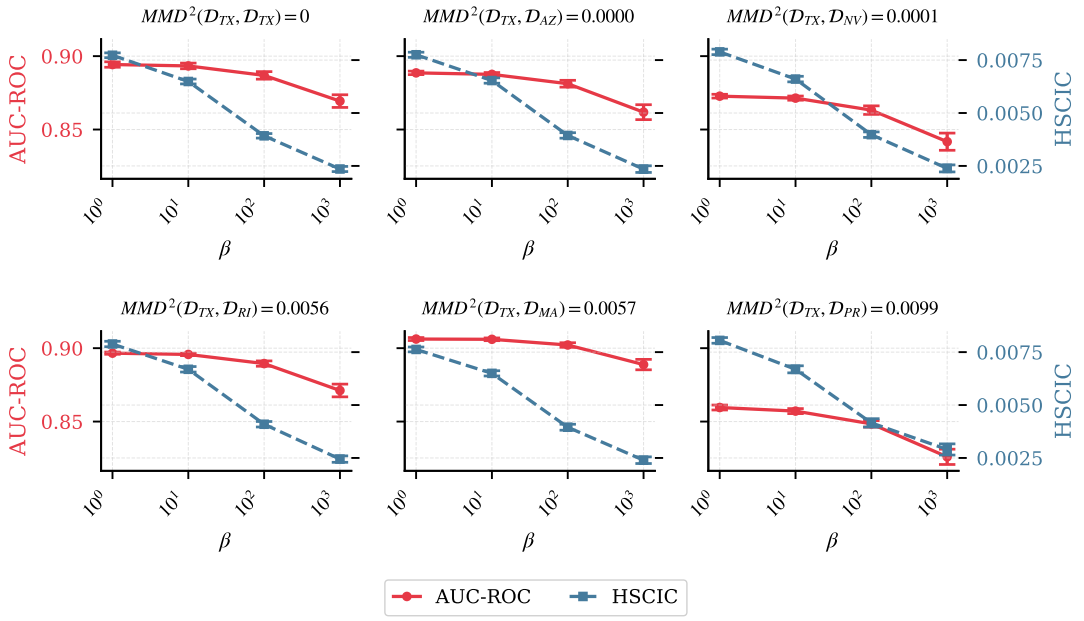


Figure A.46: Income task, fairness CIP features, TX training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

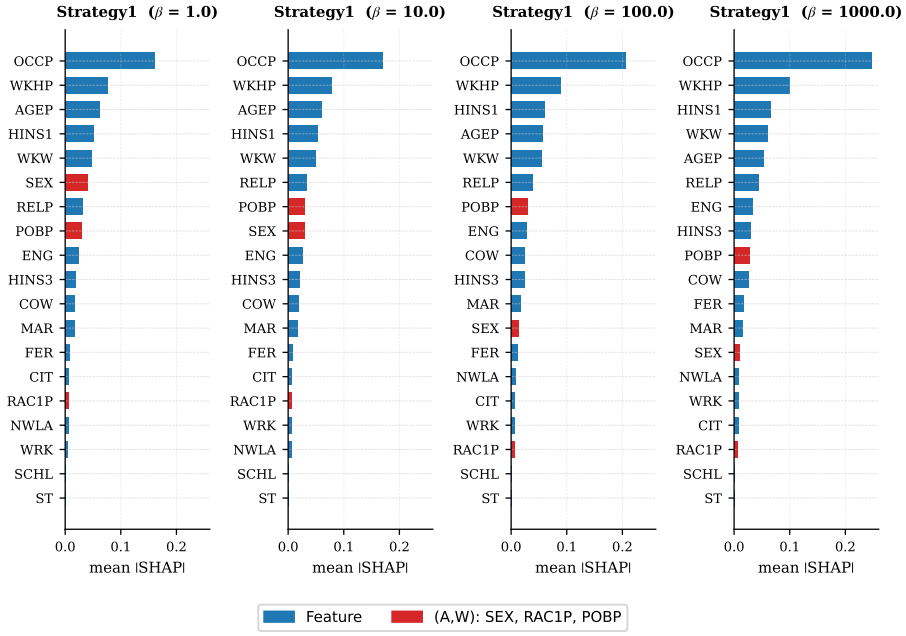


Figure A.47: SHAP analysis: income task, fairness CIP features, TX train domain (out-of-domain), STRATEGY 1 (joint optimisation). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

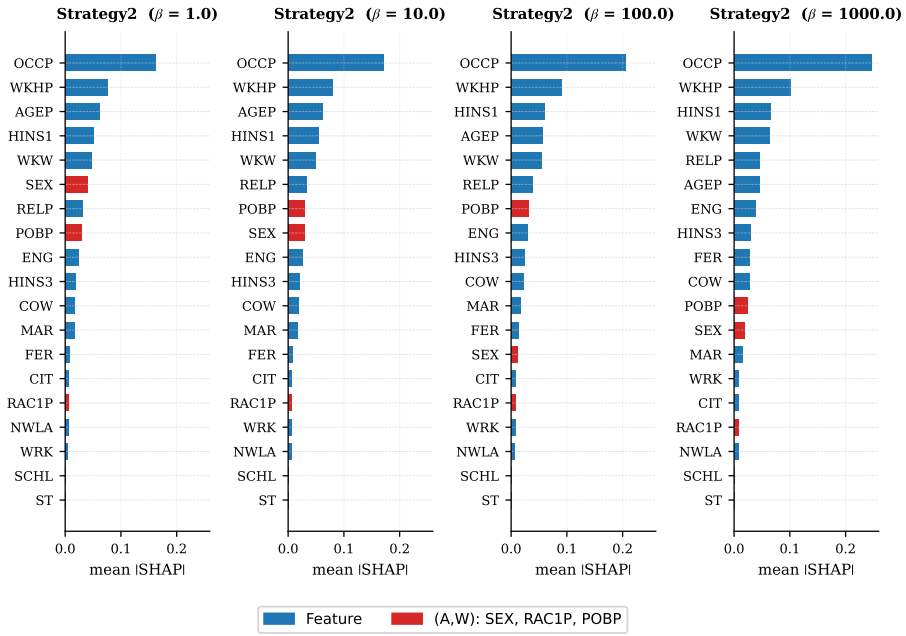


Figure A.48: SHAP analysis: income task, fairness CIP features, TX train domain (out-of-domain), STRATEGY 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

A.4 Public Coverage Task – Arguably Causal

A.4.1 CA Train Domain

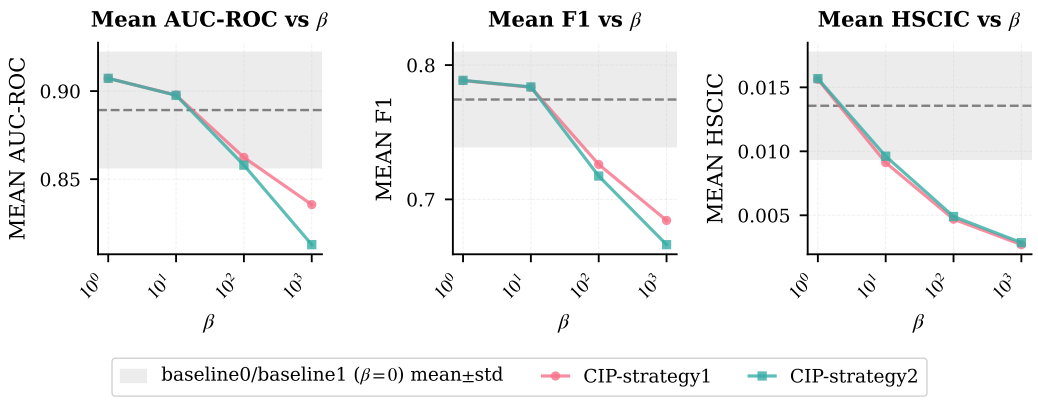


Figure A.49: Public coverage task, arguably causal CIP features, CA training domain: mean AUC-ROC, Macro-F1 and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

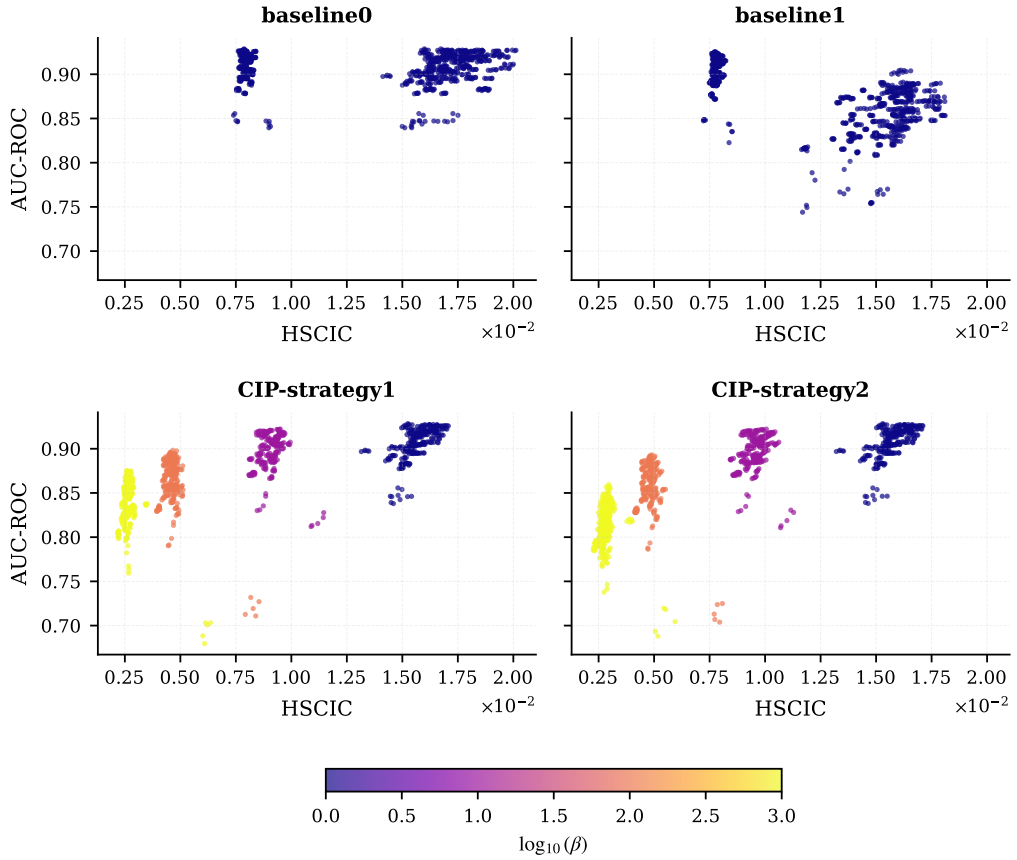


Figure A.50: Public coverage task, arguably causal CIP features, CA training domain: relation between HSCIC and AUC-ROC across baseline and CIP strategies.

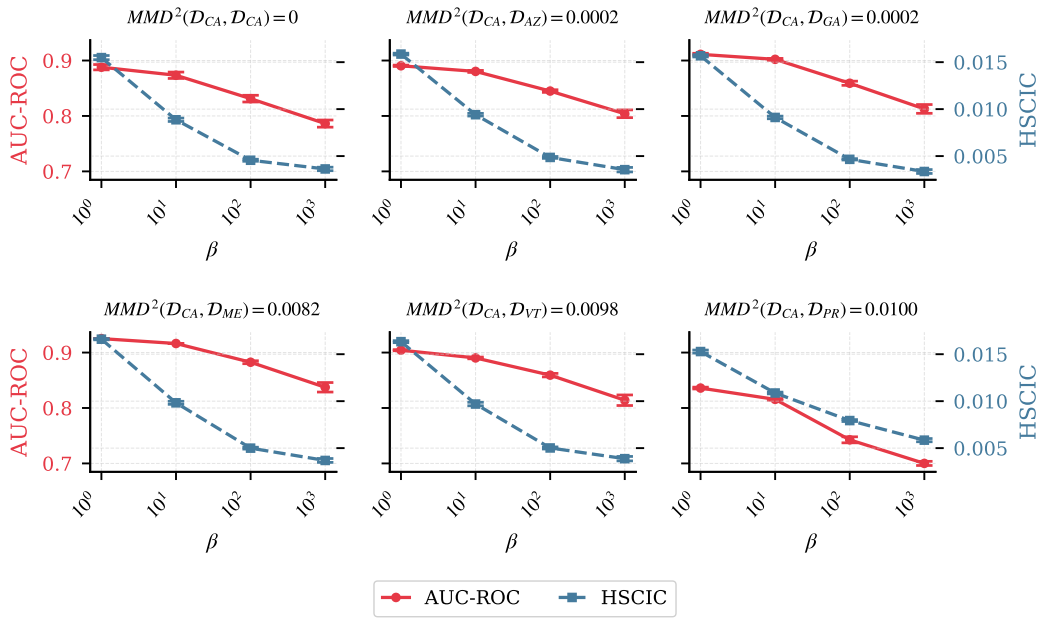


Figure A.51: Public coverage task, arguably causal CIP features, CA training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

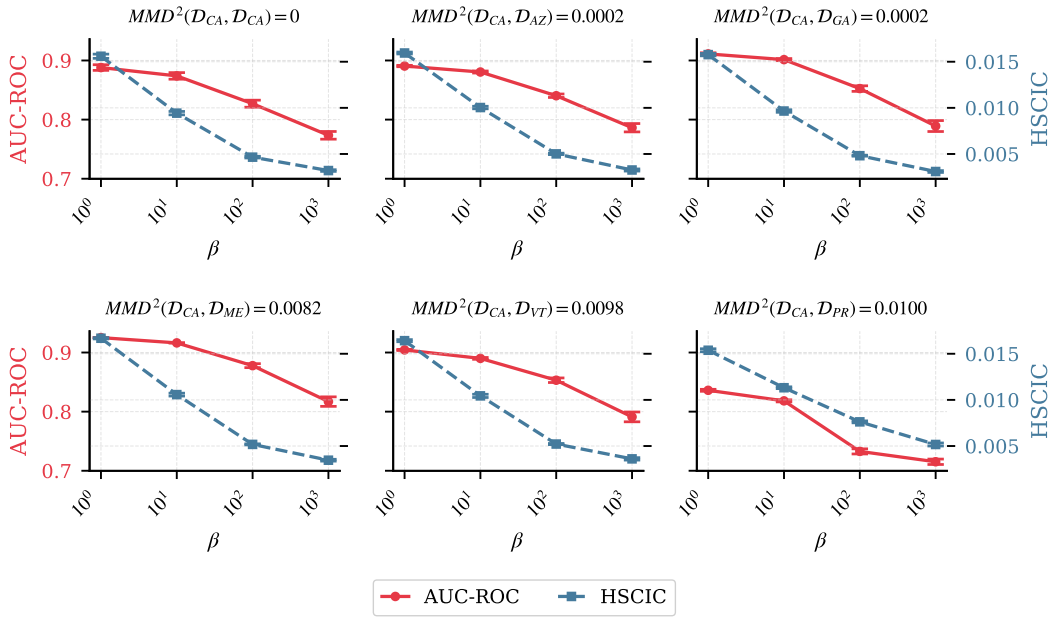


Figure A.52: Public coverage task, arguably causal CIP features, CA training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

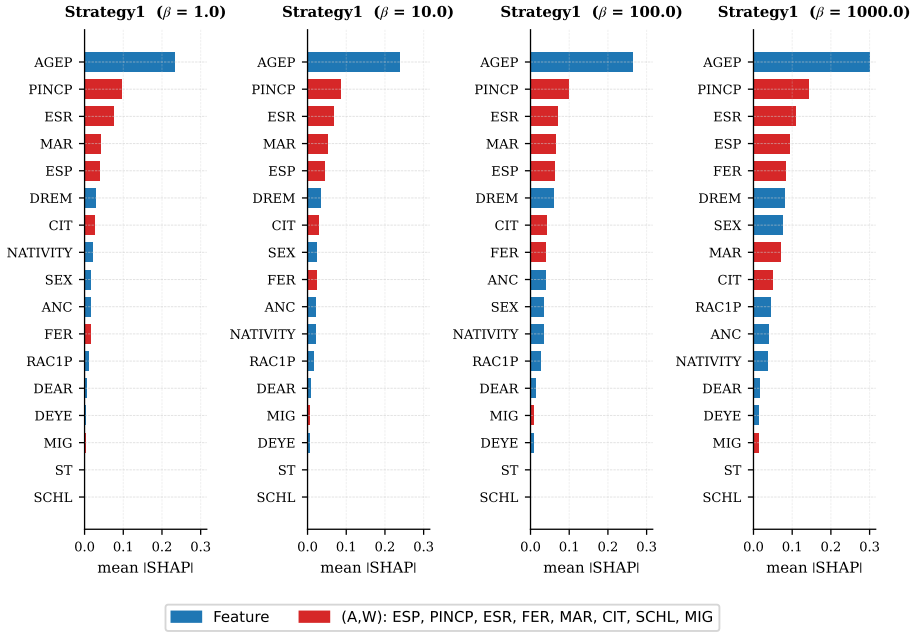


Figure A.53: SHAP analysis: public coverage task, arguably causal CIP features, CA train domain (out-of-domain), STRATEGY 1 (joint optimisation). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

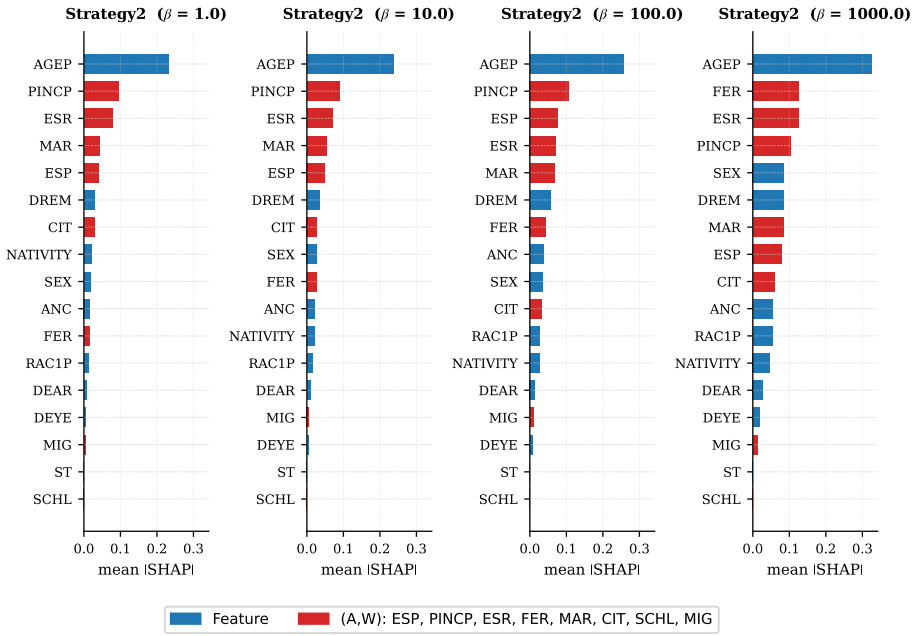


Figure A.54: SHAP analysis: public coverage task, arguably causal CIP features, CA train domain (out-of-domain), STRATEGY 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

A.4.2 MI Train Domain

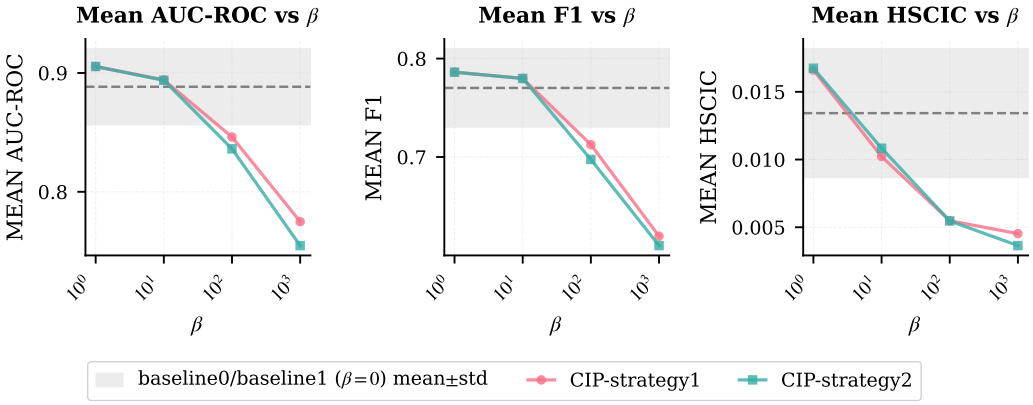


Figure A.55: Public coverage task, arguably causal CIP features, MI training domain: mean AUC-ROC, Macro-F1 and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

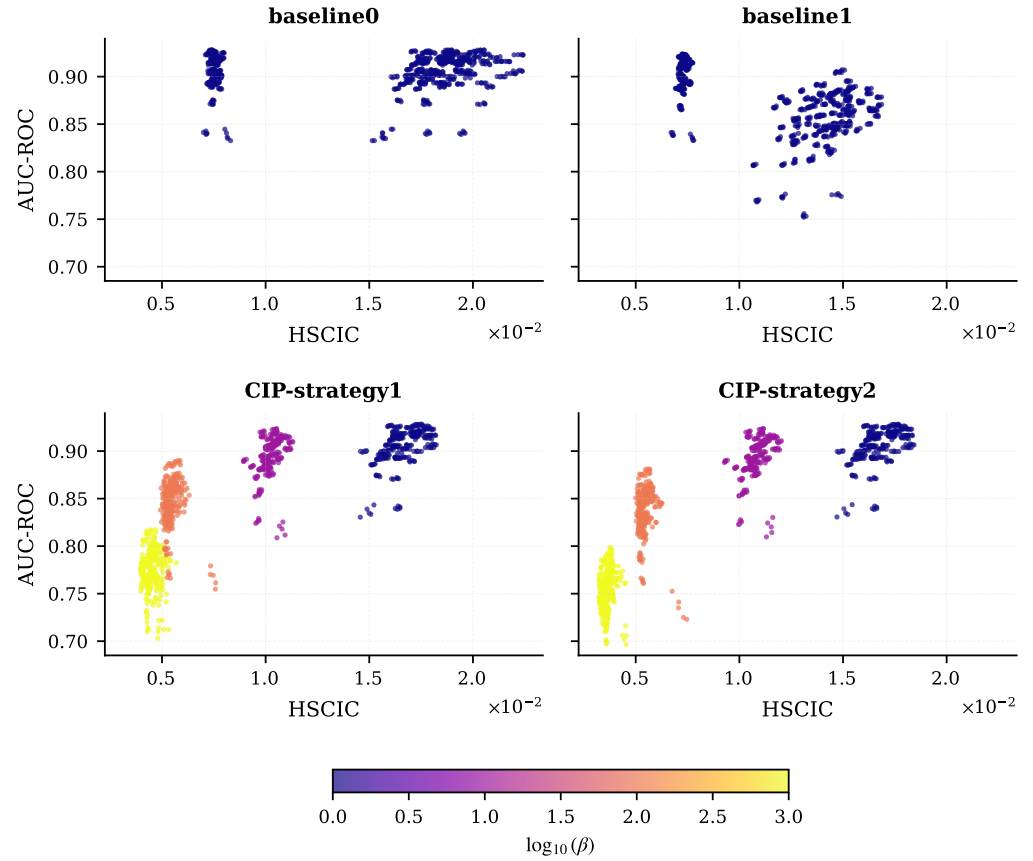


Figure A.56: Public coverage task, arguably causal CIP features, MI training domain: relation between HSCIC and AUC-ROC across baseline and CIP strategies.

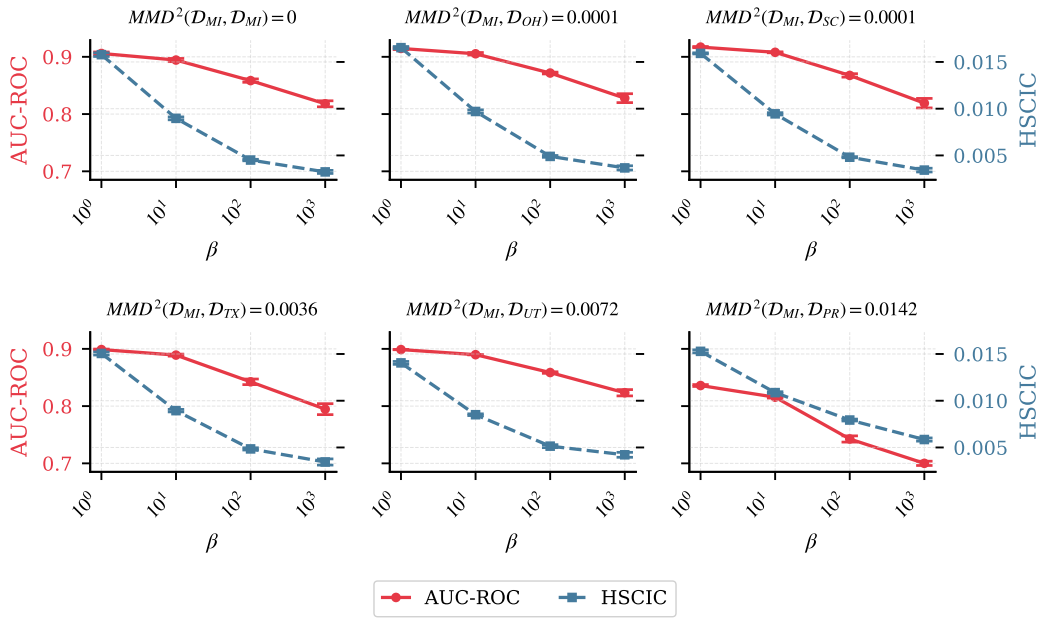


Figure A.57: Public coverage task, arguably causal CIP features, MI training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

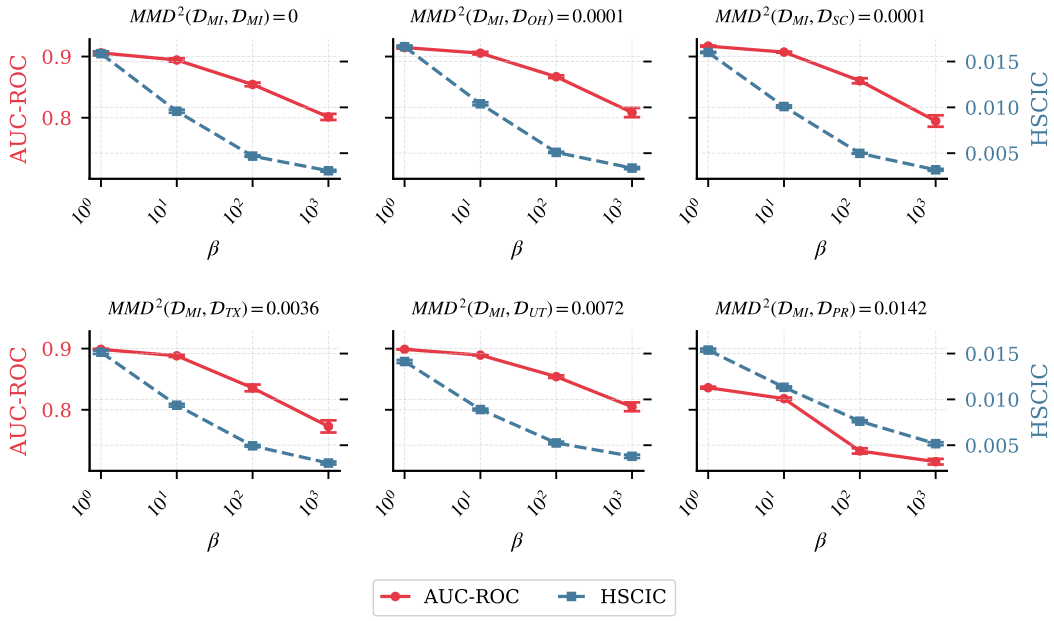


Figure A.58: Public coverage task, arguably causal CIP features, MI training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

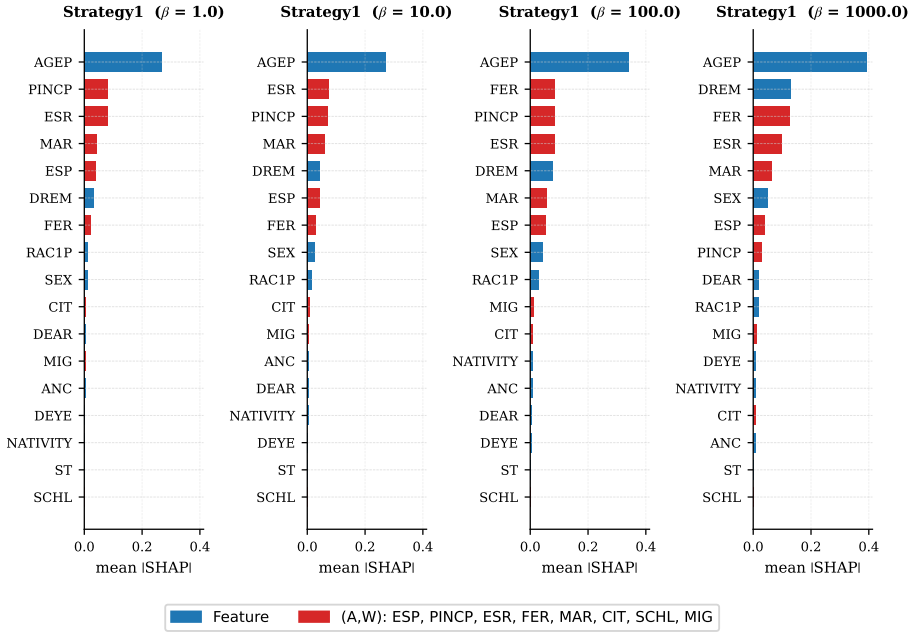


Figure A.59: SHAP analysis: public coverage task, arguably causal CIP features, MI train domain (out-of-domain), Strategy 1 (joint optimisation). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

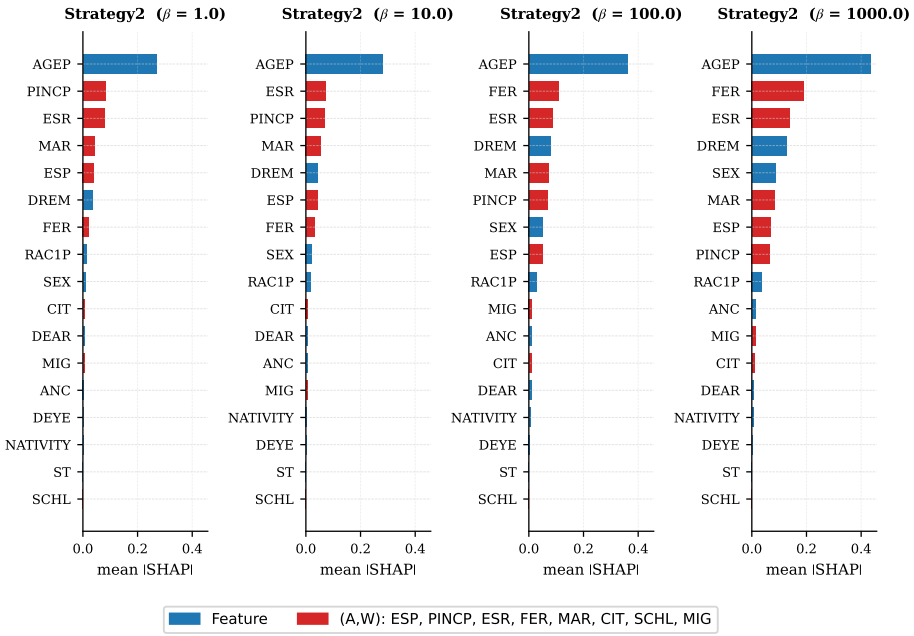


Figure A.60: SHAP analysis: public coverage task, arguably causal CIP features, MI train domain (out-of-domain), Strategy 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

A.4.3 TX Train Domain

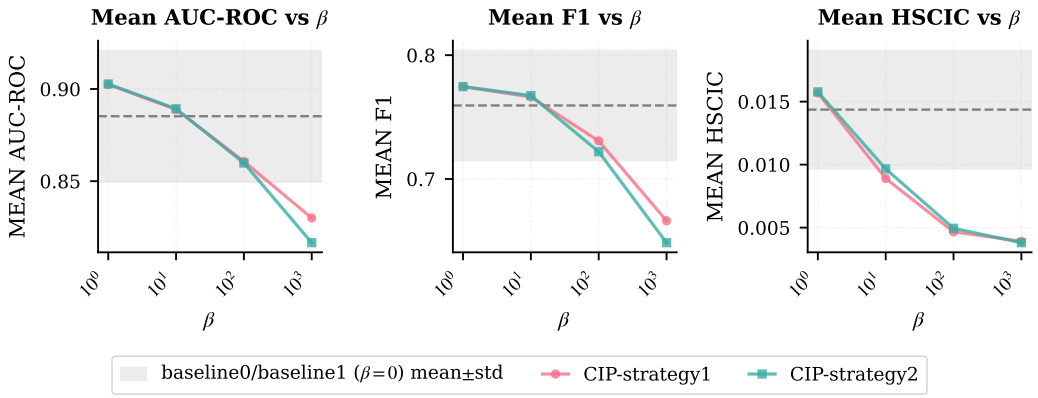


Figure A.61: Public coverage task, arguably causal CIP features, TX training domain: mean AUC-ROC, Macro-F1 and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

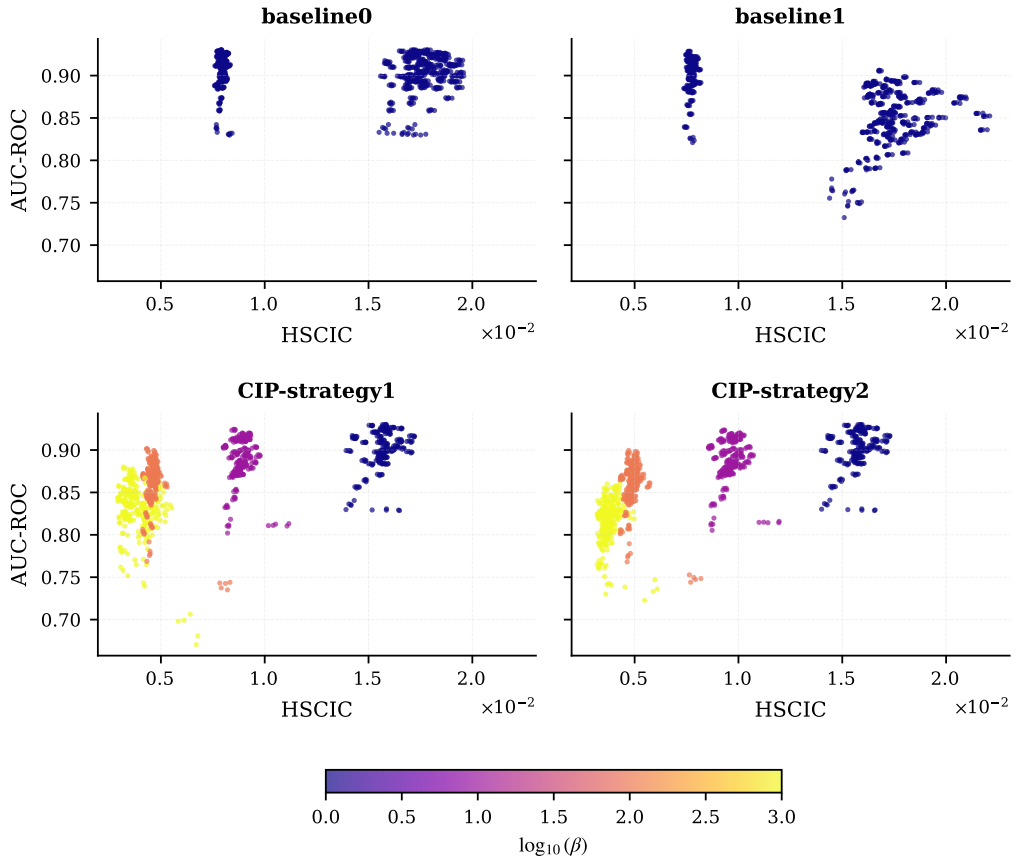


Figure A.62: Public coverage task, arguably causal CIP features, TX training domain: relation between HSCIC and AUC-ROC across baseline and CIP strategies.

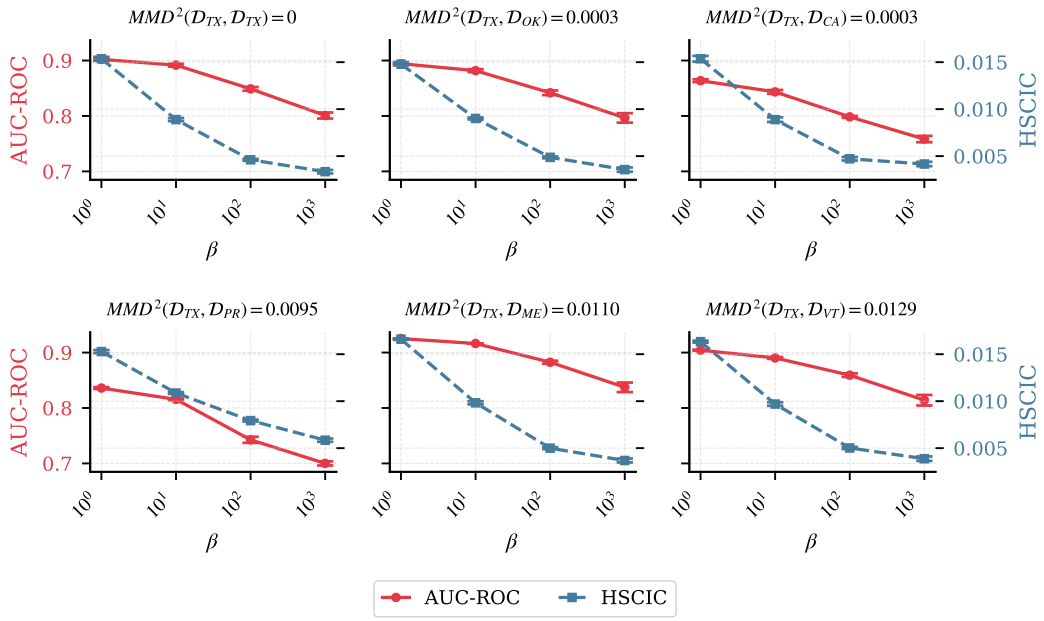


Figure A.63: Public coverage task, arguably causal CIP features, TX training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

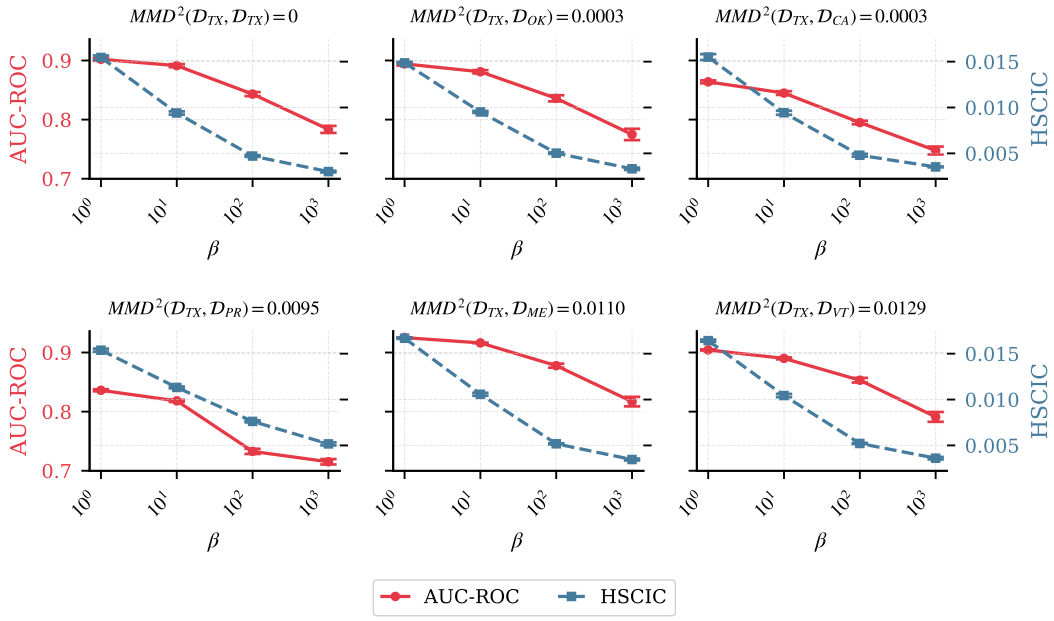


Figure A.64: Public coverage task, arguably causal CIP features, TX training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

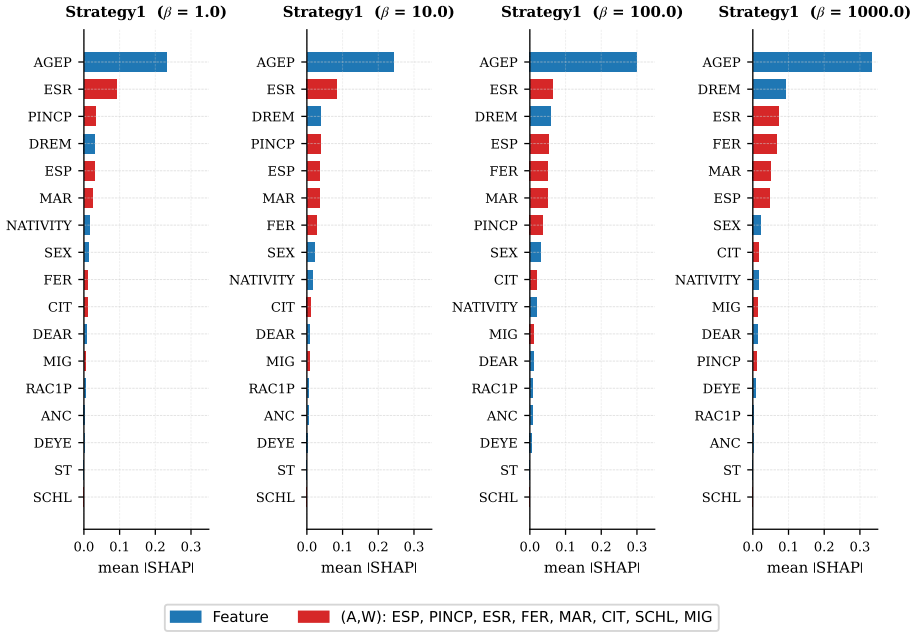


Figure A.65: SHAP analysis: public coverage task, arguably causal CIP features, TX train domain (out-of-domain), Strategy 1 (joint optimisation). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

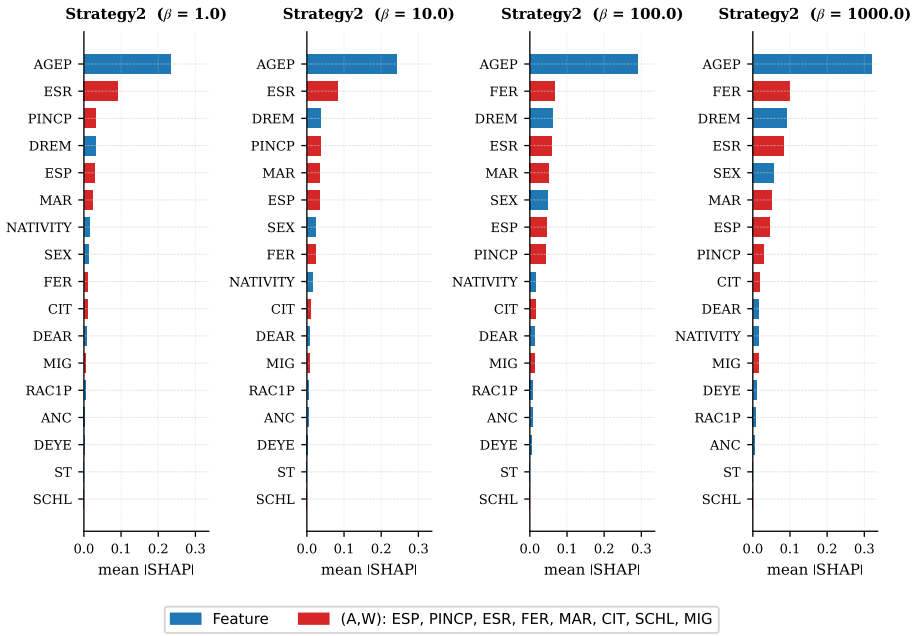


Figure A.66: SHAP analysis: public coverage task, arguably causal CIP features, TX train domain (out-of-domain), Strategy 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

A.5 Public Coverage Task – State Dependent

A.5.1 CA Train Domain

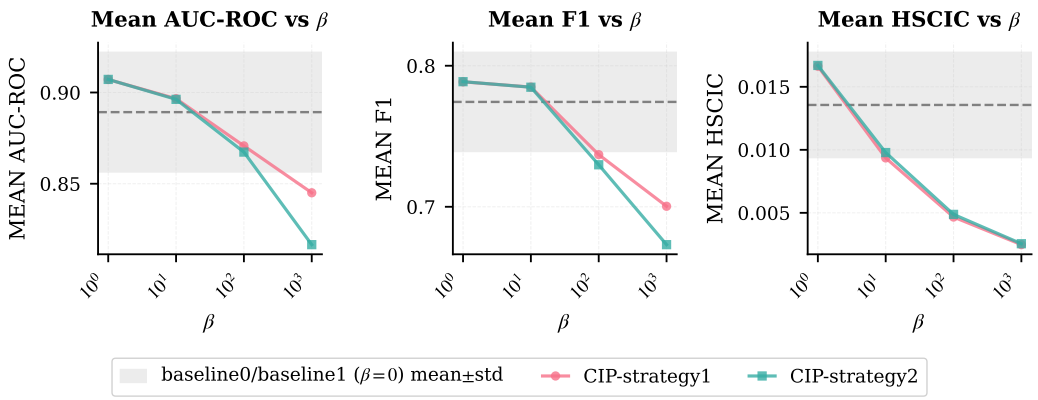


Figure A.67: Public coverage task, state-dependent CIP features, CA training domain: mean AUC-ROC, Macro-F1 and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

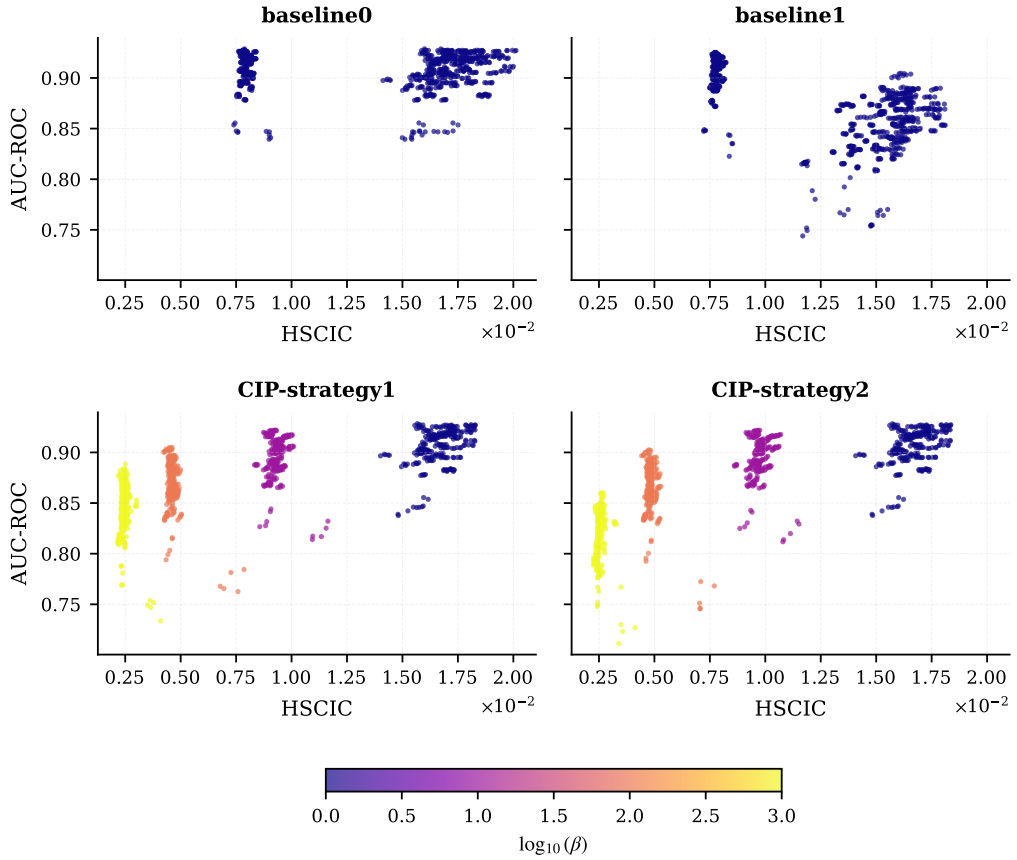


Figure A.68: Public coverage task, state-dependent CIP features, CA training domain: relation between HSCIC and AUC-ROC across baseline and CIP strategies.

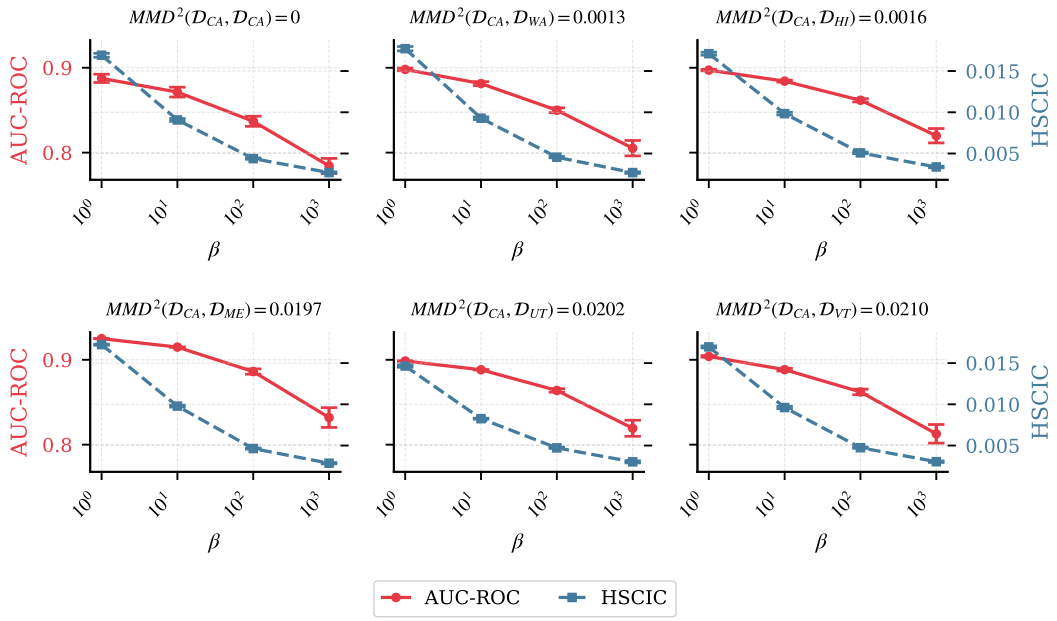


Figure A.69: Public coverage task, state-dependent CIP features, CA training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

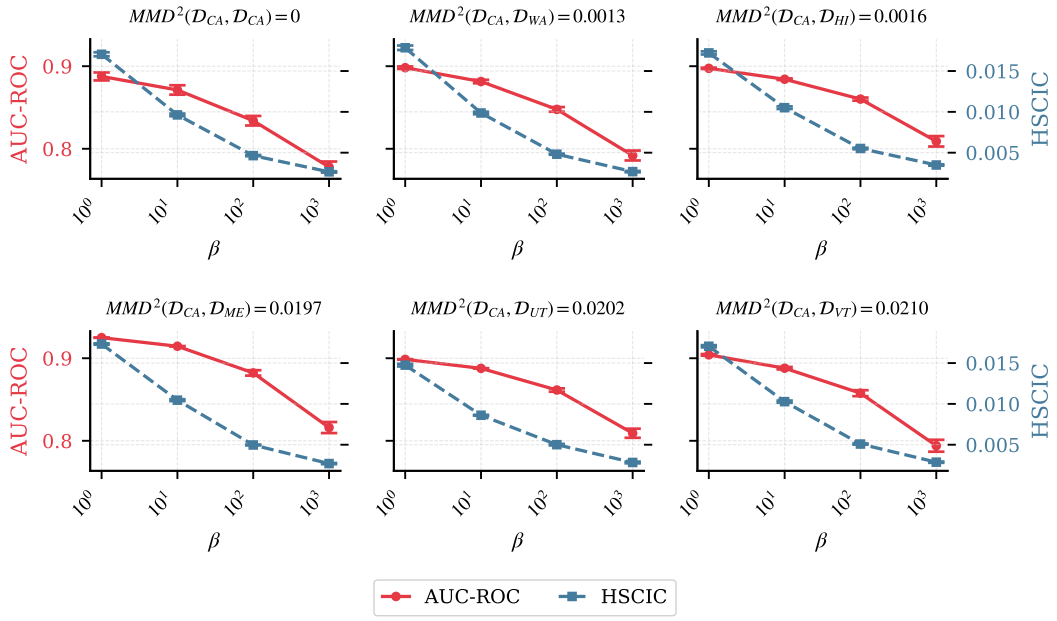


Figure A.70: Public coverage task, state-dependent CIP features, CA training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

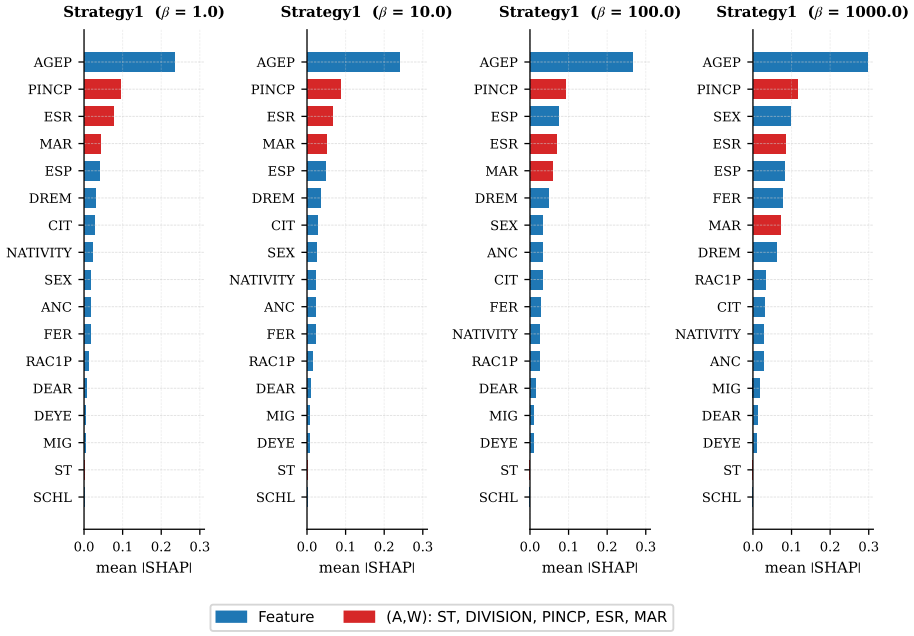


Figure A.71: SHAP analysis: public coverage task, state-dependent CIP features, CA train domain (out-of-domain), STRATEGY 1 (joint optimisation). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

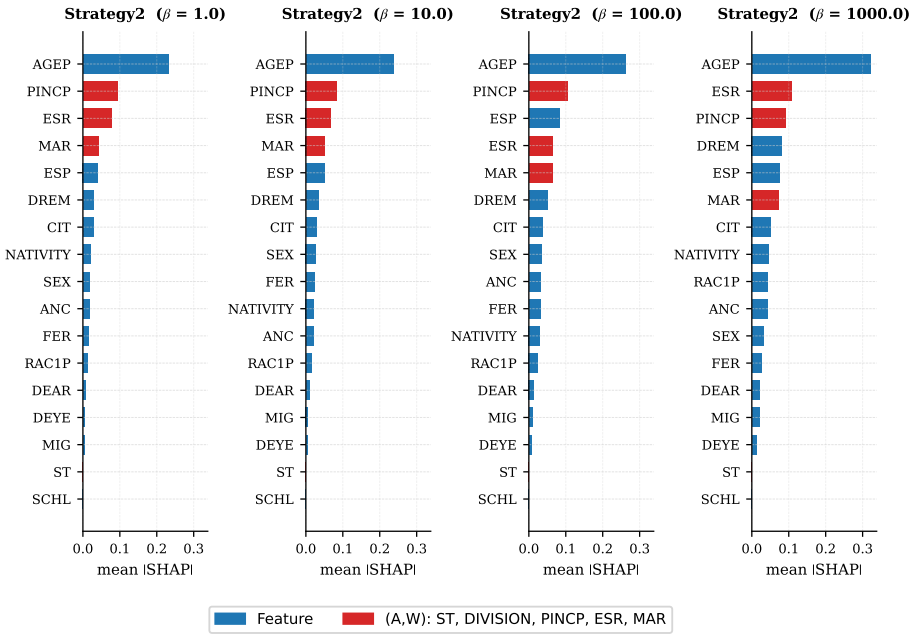


Figure A.72: SHAP analysis: public coverage task, state-dependent CIP features, CA train domain (out-of-domain), STRATEGY 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

A.5.2 MI Train Domain

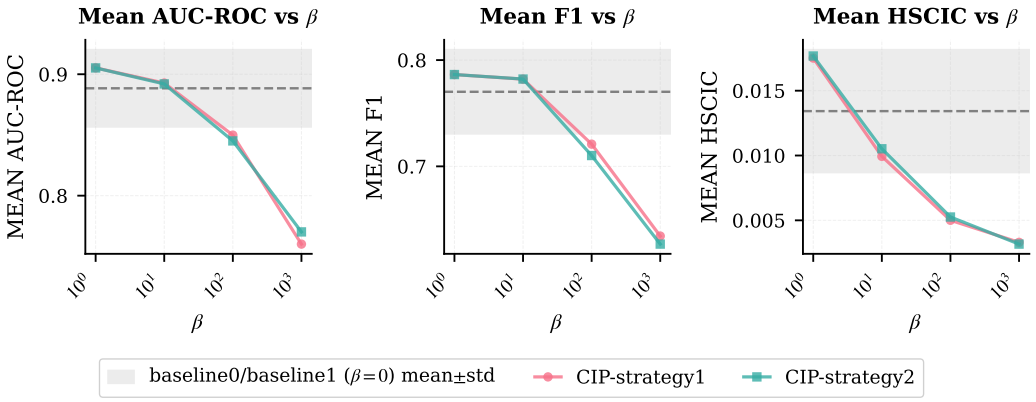


Figure A.73: Public coverage task, state-dependent CIP features, MI training domain: mean AUC-ROC, Macro-F1 and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

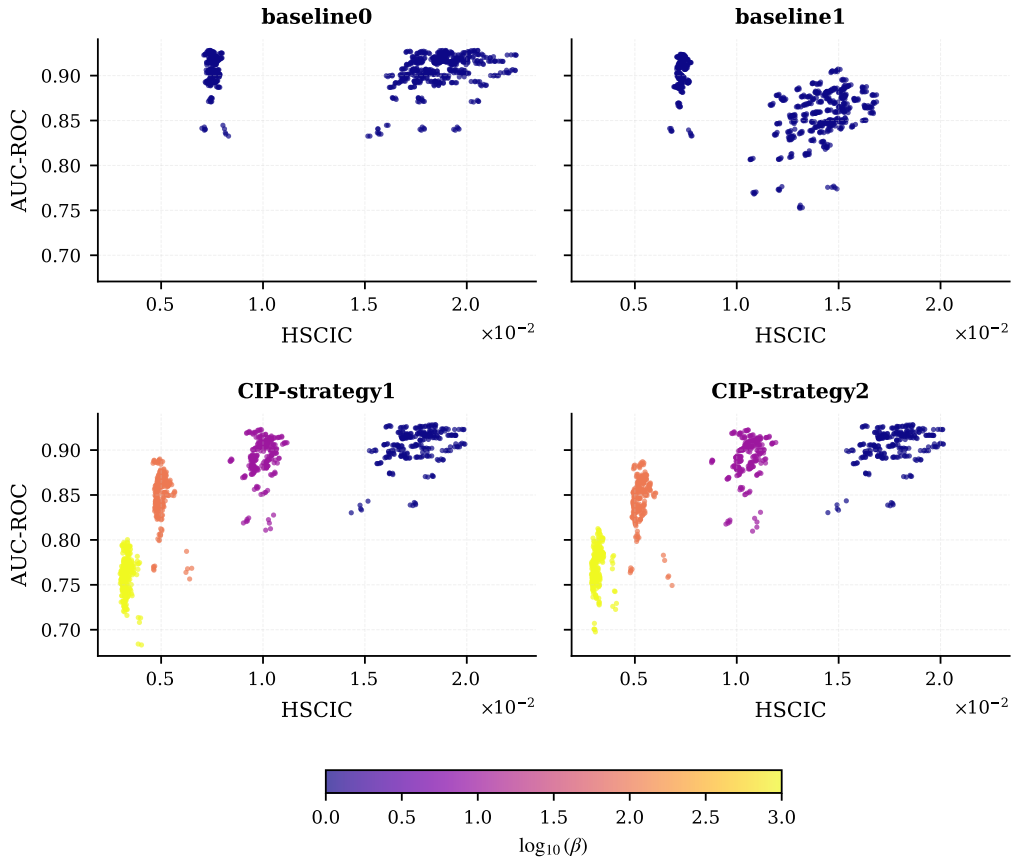


Figure A.74: Public coverage task, state-dependent CIP features, MI training domain: relation between HSCIC and AUC-ROC across baseline and CIP strategies.

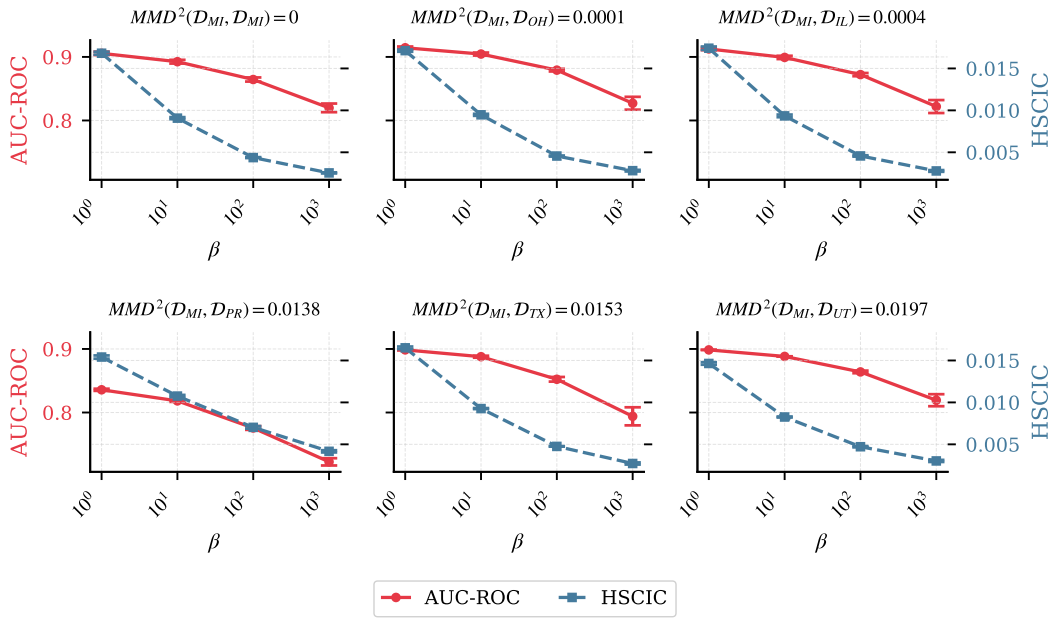


Figure A.75: Public coverage task, state-dependent CIP features, MI training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

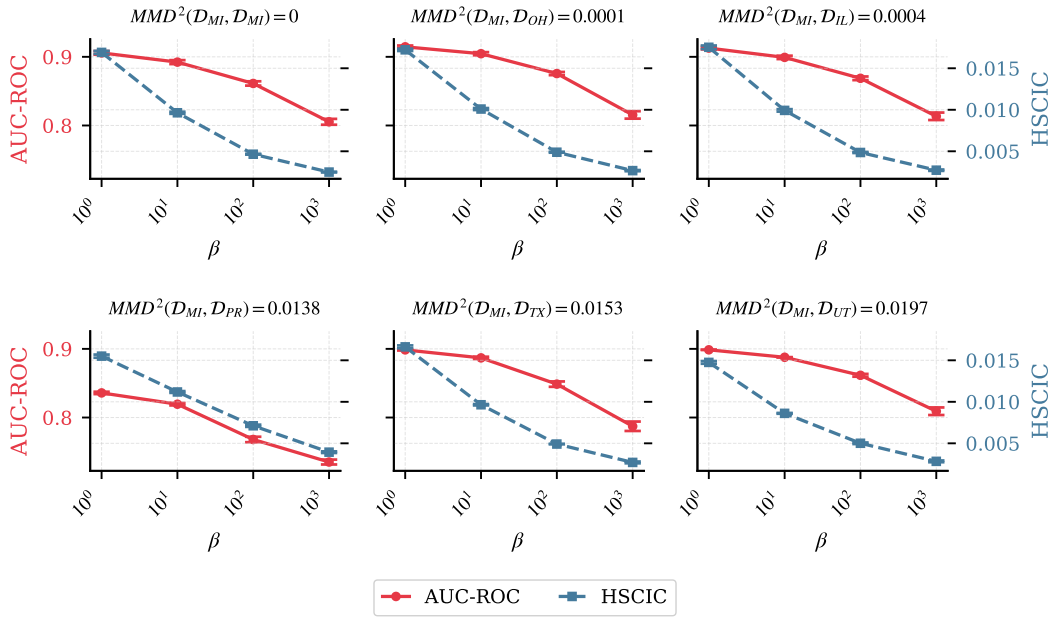


Figure A.76: Public coverage task, state-dependent CIP features, MI training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

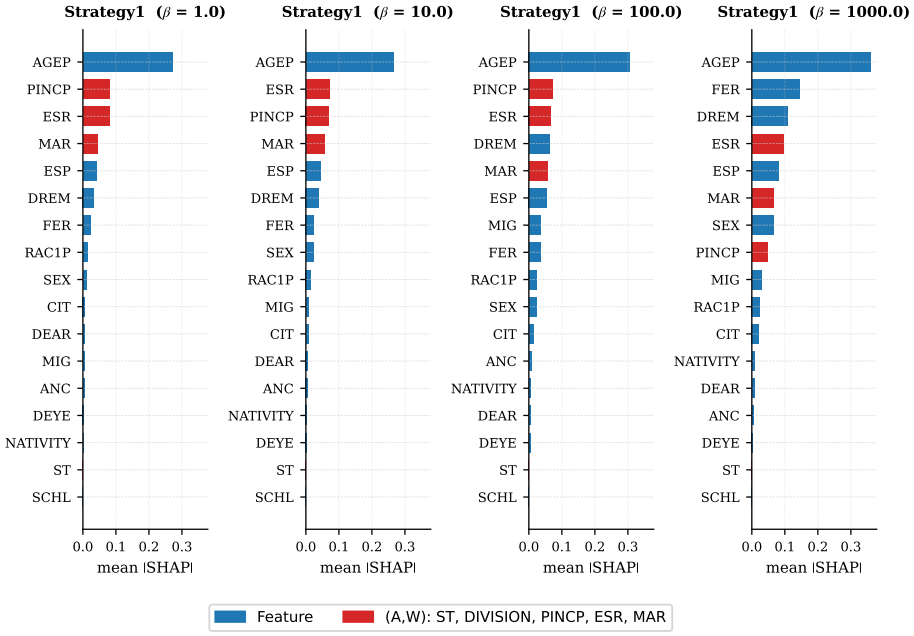


Figure A.77: SHAP analysis: public coverage task, state-dependent CIP features, MI train domain (out-of-domain), STRATEGY 1 (joint optimisation). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

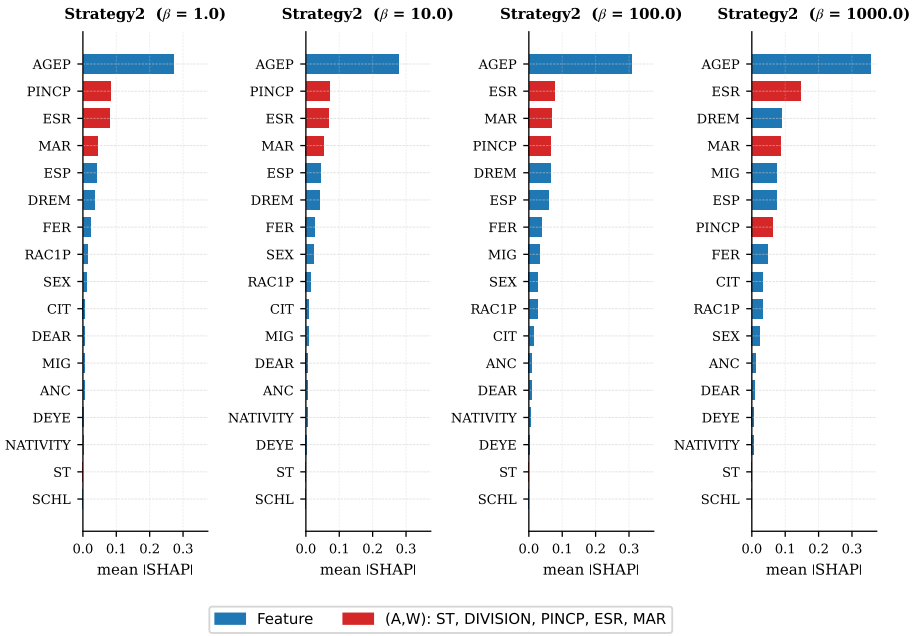


Figure A.78: SHAP analysis: public coverage task, state-dependent CIP features, MI train domain (out-of-domain), STRATEGY 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

A.5.3 TX Train Domain

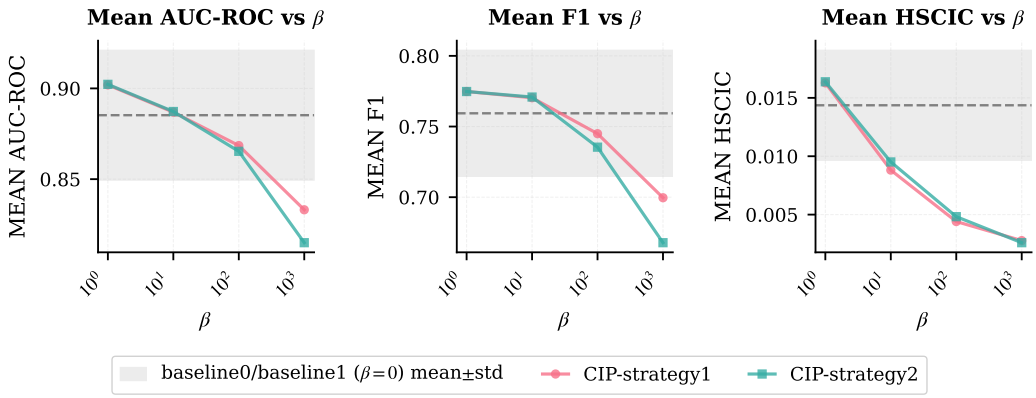


Figure A.79: Public coverage task, state-dependent CIP features, TX training domain: mean AUC-ROC, Macro-F1 and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

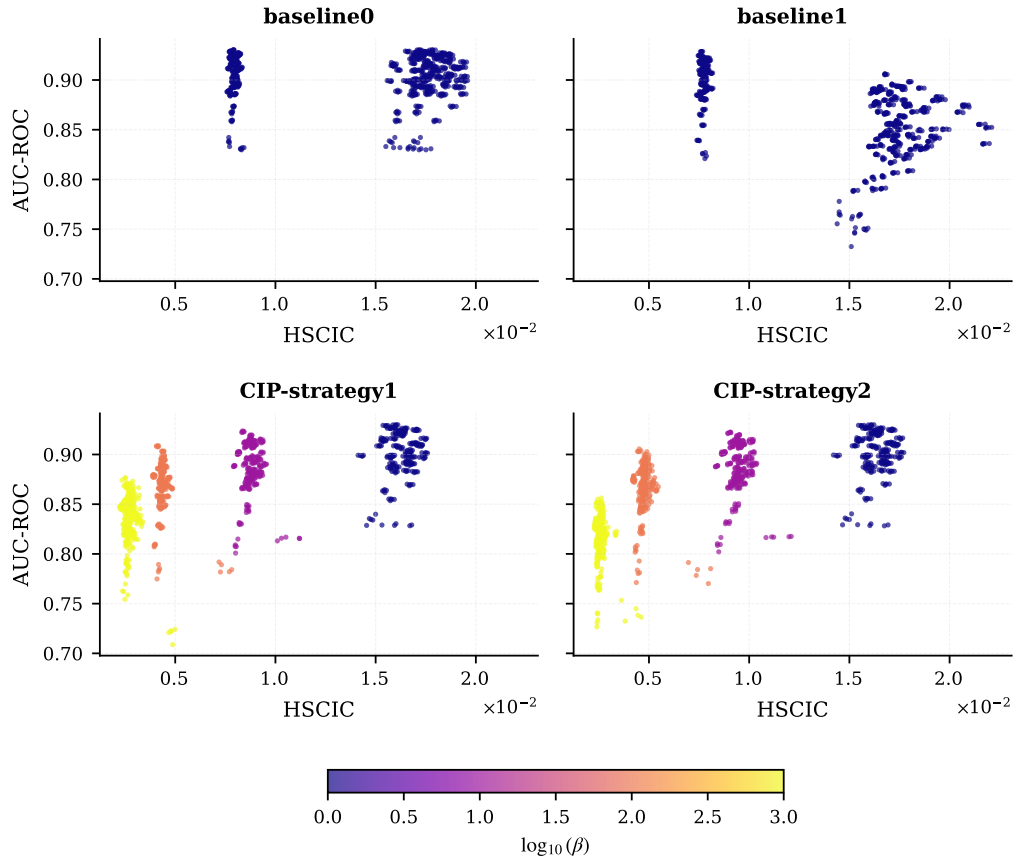


Figure A.80: Public coverage task, state-dependent CIP features, TX training domain: relation between HSCIC and AUC-ROC across baseline and CIP strategies.

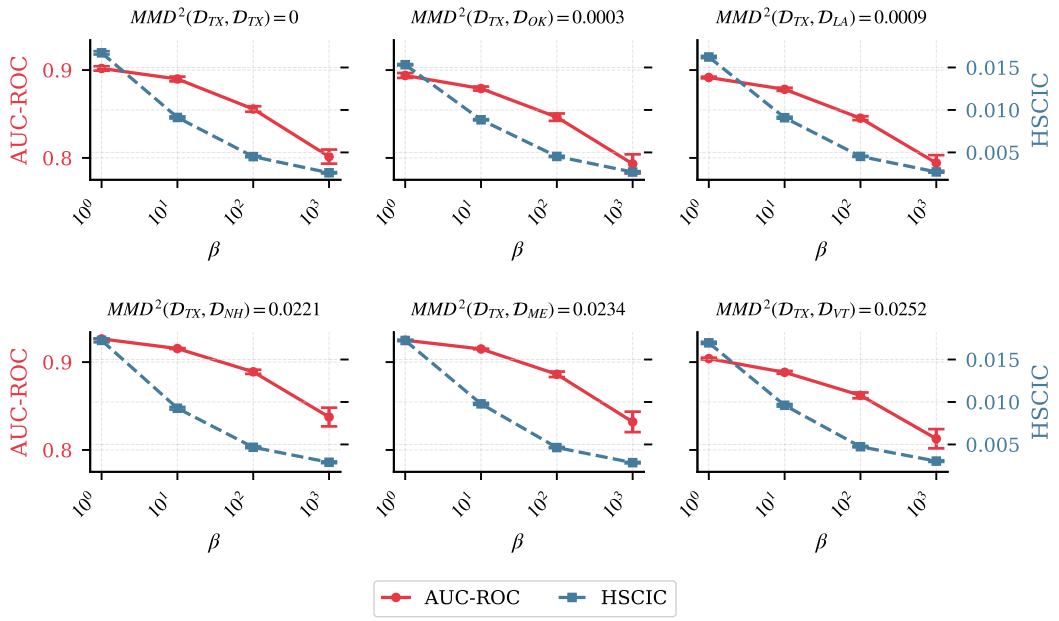


Figure A.81: Public coverage task, state-dependent CIP features, TX training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

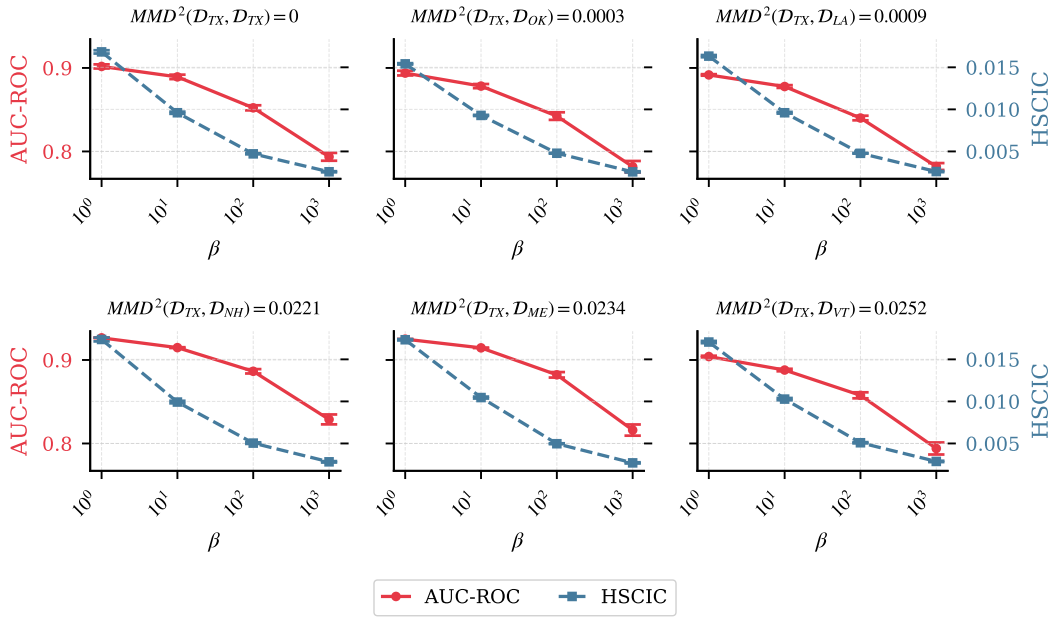


Figure A.82: Public coverage task, state-dependent CIP features, TX training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

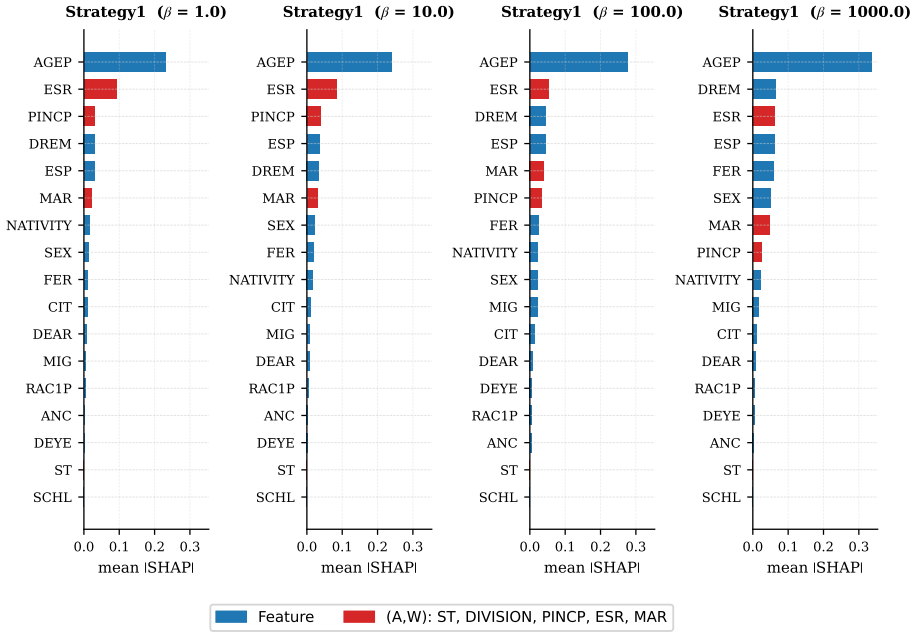


Figure A.83: SHAP analysis: public coverage task, state-dependent CIP features, TX train domain (out-of-domain), STRATEGY 1 (joint optimisation). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

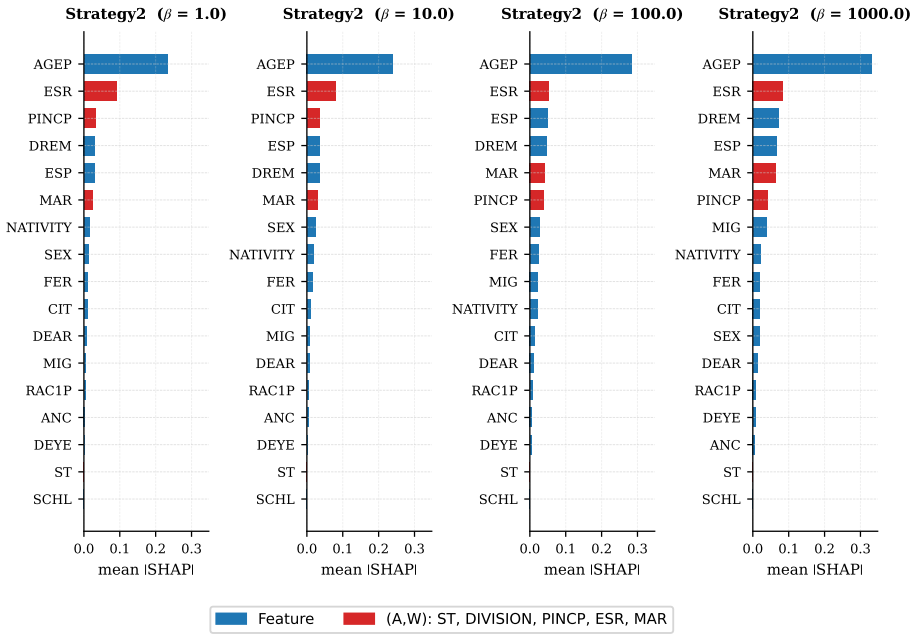


Figure A.84: SHAP analysis: public coverage task, state-dependent CIP features, TX train domain (out-of-domain), STRATEGY 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

A.6 Public Coverage Task – Fairness

A.6.1 CA Train Domain

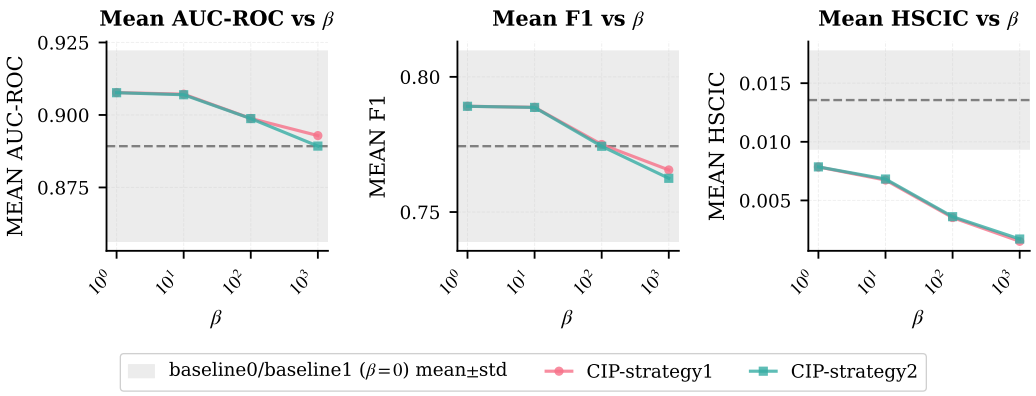


Figure A.85: Public coverage task, fairness CIP features, CA training domain: mean AUC-ROC, Macro-F1 and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

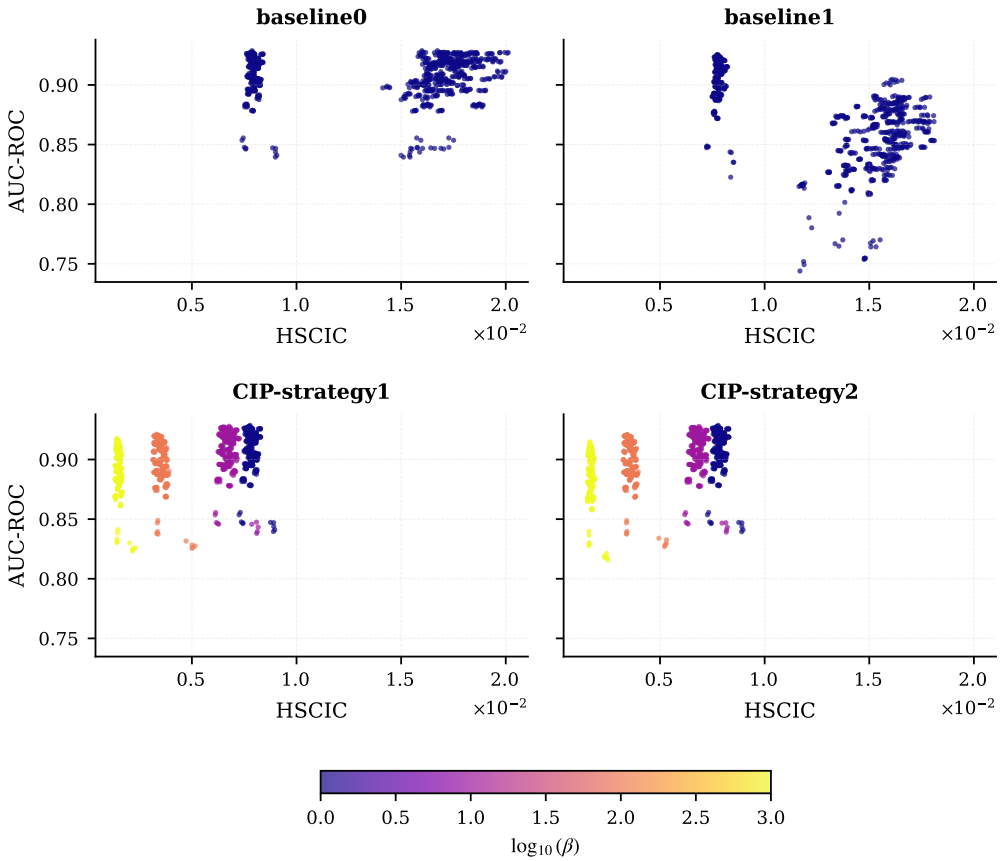


Figure A.86: Public coverage task, fairness CIP features, CA training domain: relation between HSCIC and AUC-ROC across baseline and CIP strategies.

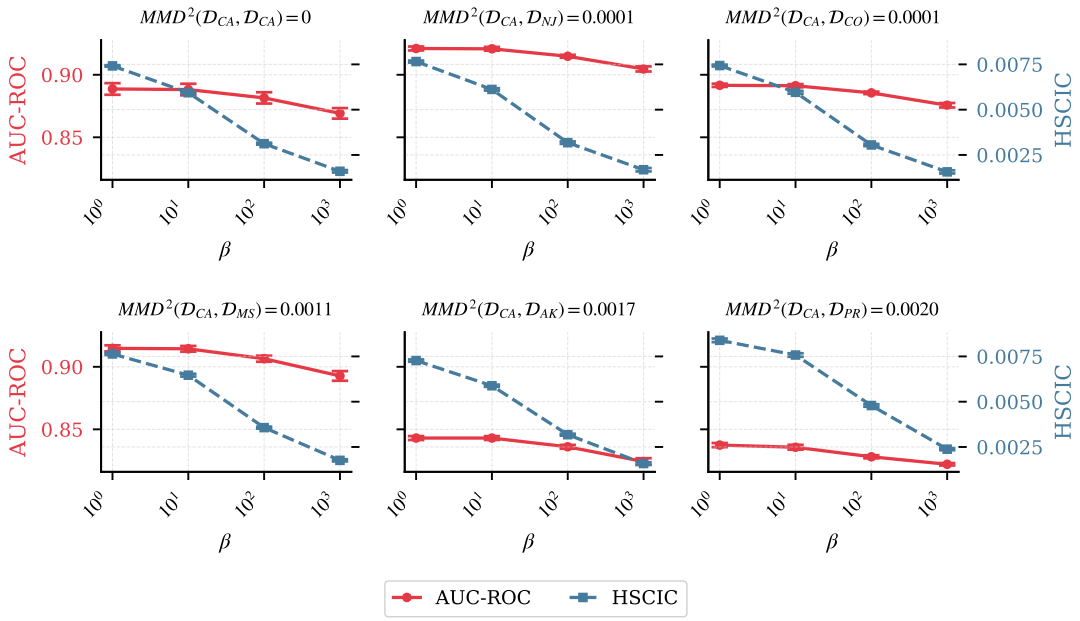


Figure A.87: Public coverage task, fairness CIP features, CA training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

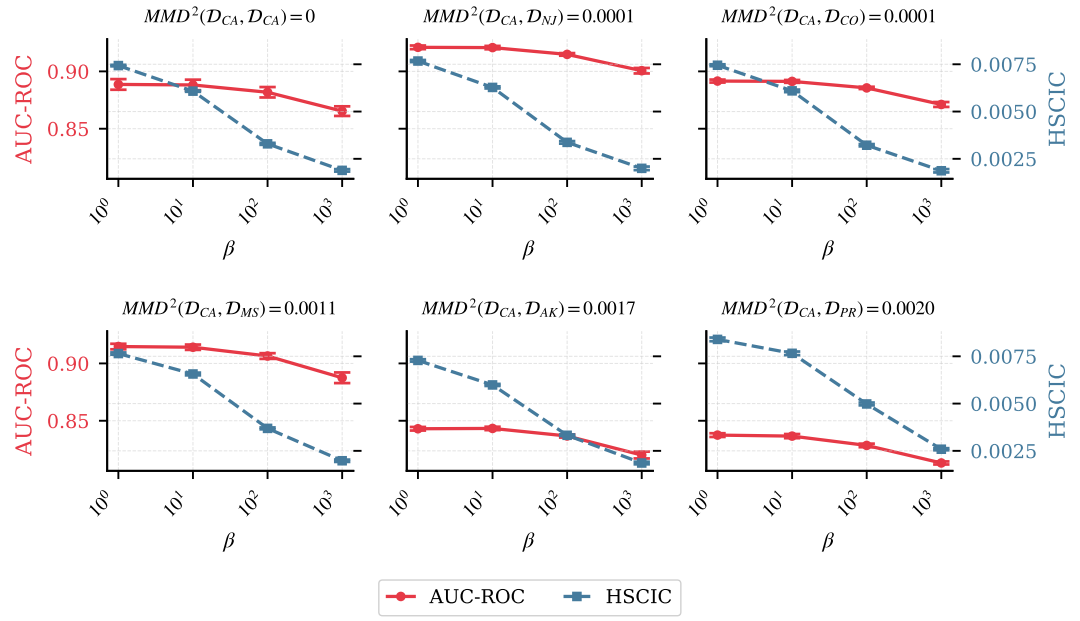


Figure A.88: Public coverage task, fairness CIP features, CA training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

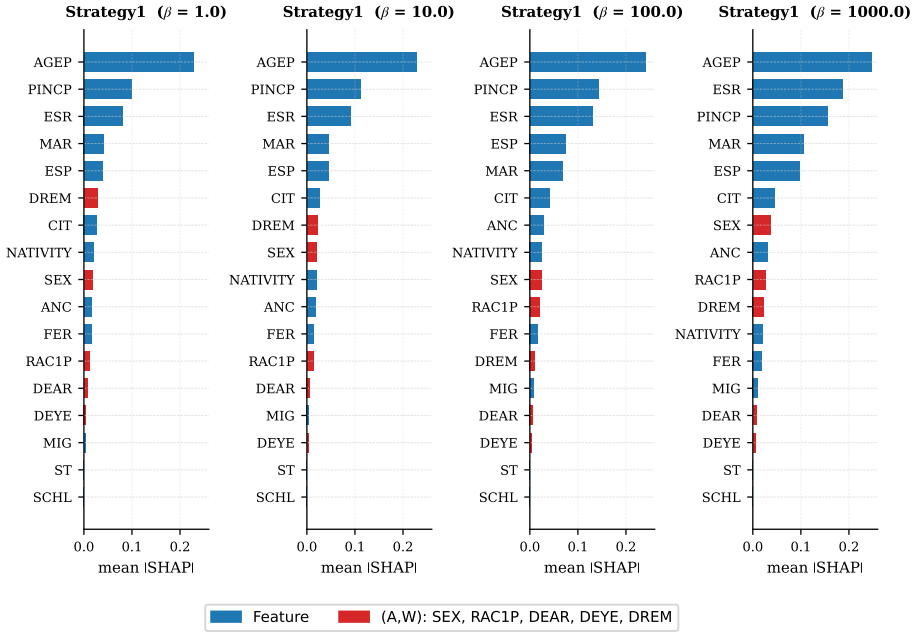


Figure A.89: SHAP analysis: public coverage task, fairness CIP features, CA train domain (out-of-domain), STRATEGY 1 (joint optimisation). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

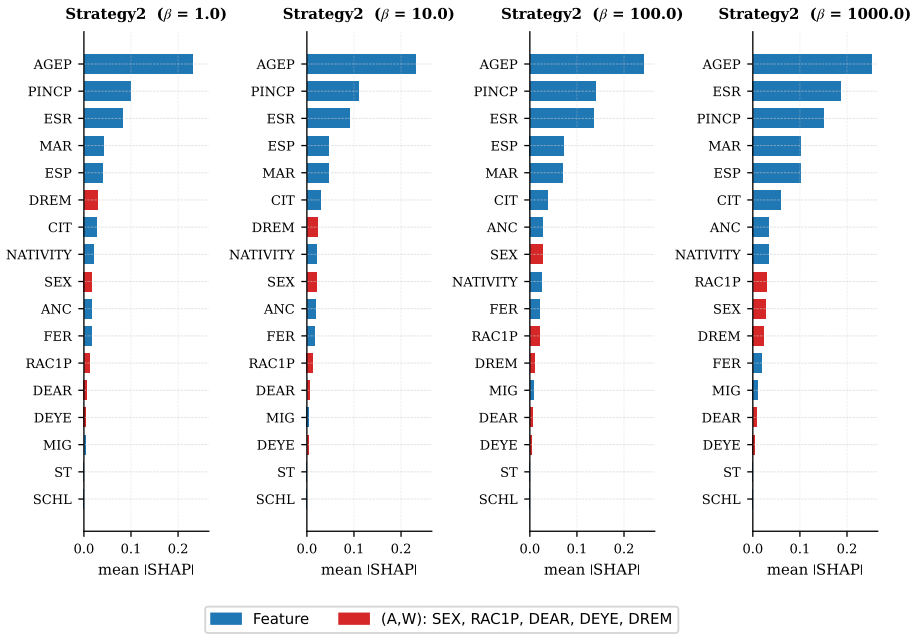


Figure A.90: SHAP analysis: public coverage task, fairness CIP features, CA train domain (out-of-domain), STRATEGY 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

A.6.2 MI Train Domain

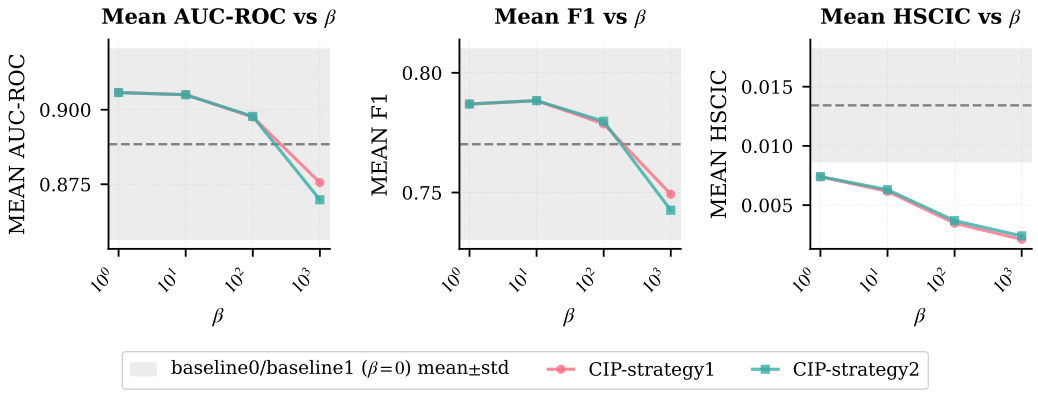


Figure A.91: Public coverage task, fairness CIP features, MI training domain: mean AUC-ROC, Macro-F1 and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

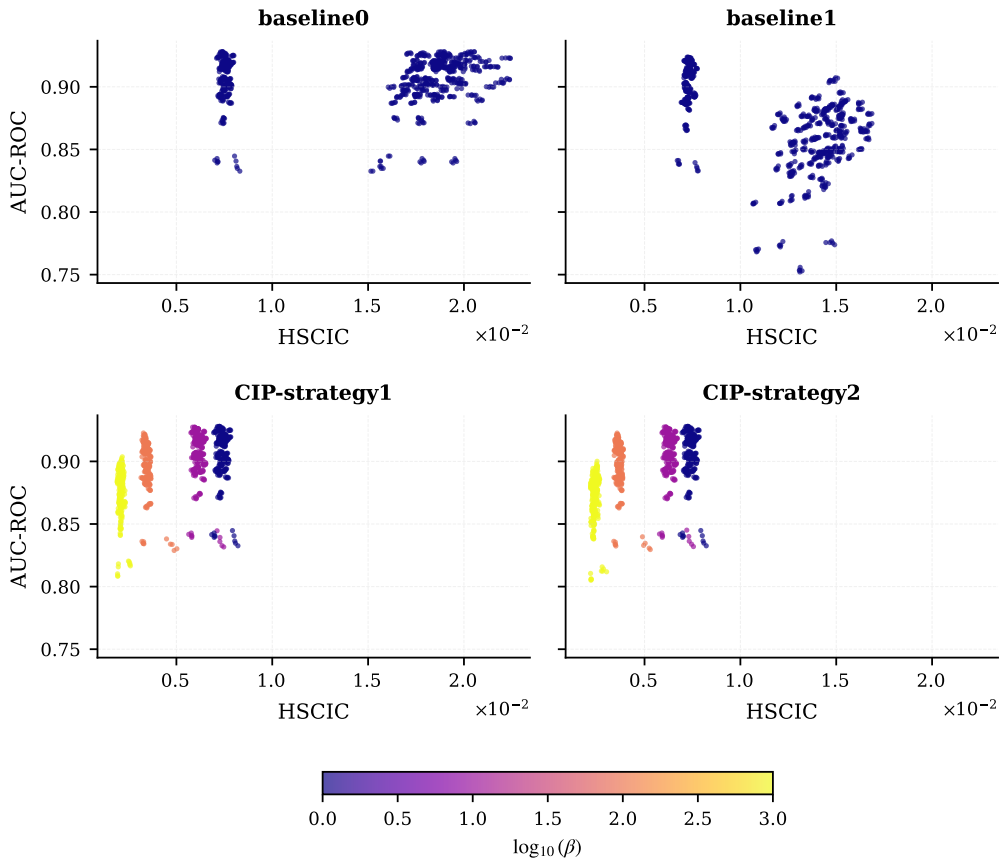


Figure A.92: Public coverage task, fairness CIP features, MI training domain: relation between HSCIC and AUC-ROC across baseline and CIP strategies.

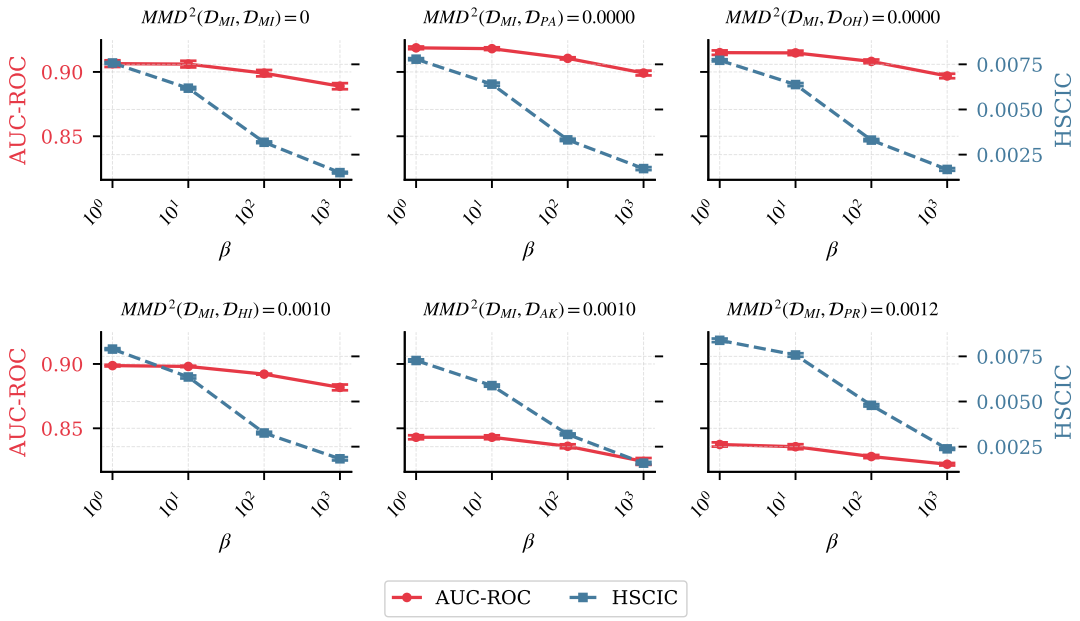


Figure A.93: Public coverage task, fairness CIP features, MI training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

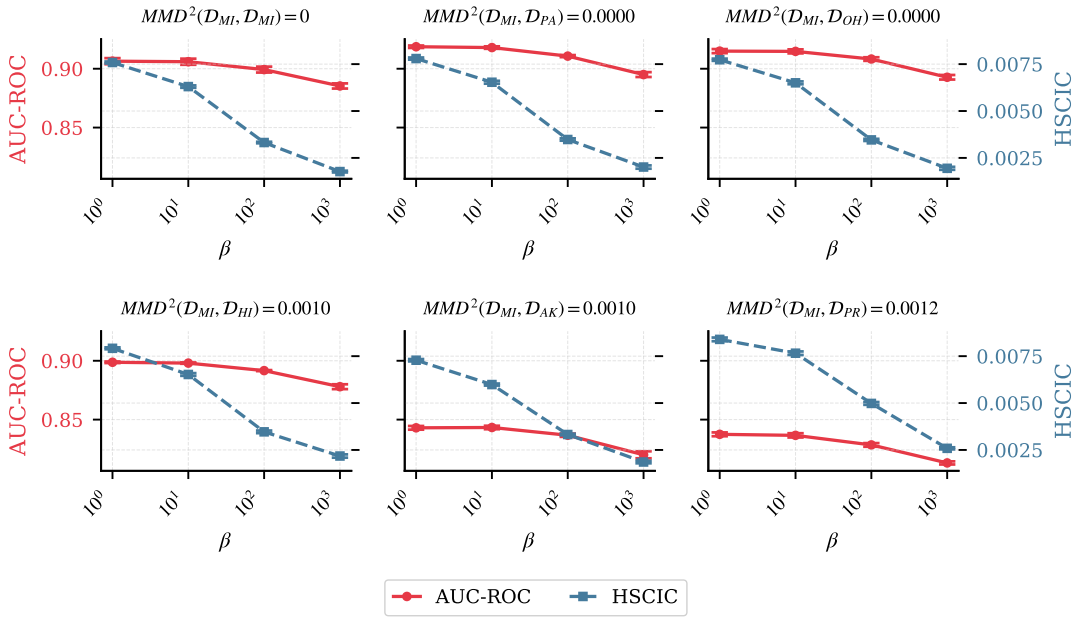


Figure A.94: Public coverage task, fairness CIP features, MI training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

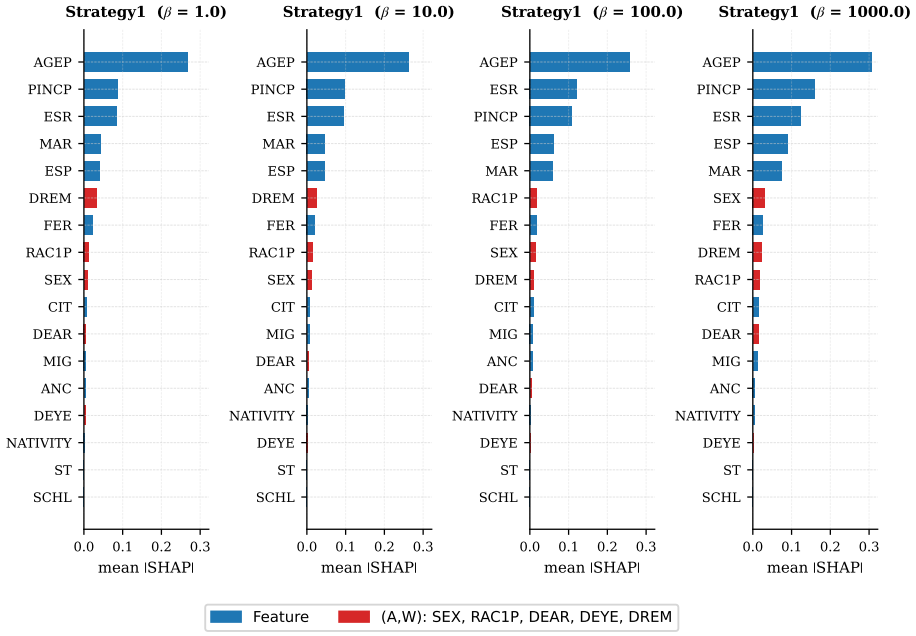


Figure A.95: SHAP analysis: public coverage task, fairness CIP features, MI train domain (out-of-domain), Strategy 1 (joint optimisation). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

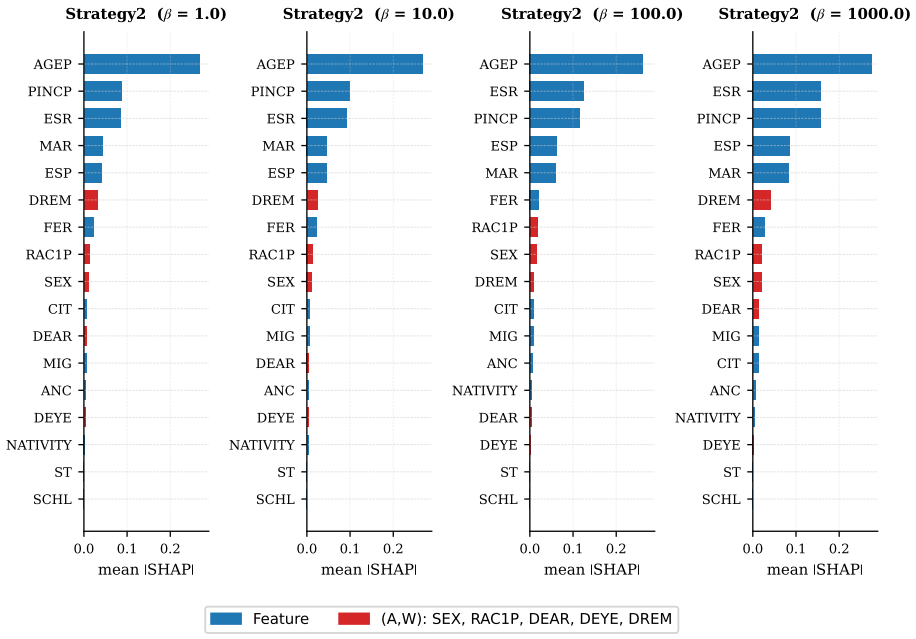


Figure A.96: SHAP analysis: public coverage task, fairness CIP features, MI train domain (out-of-domain), Strategy 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

A.6.3 TX Train Domain

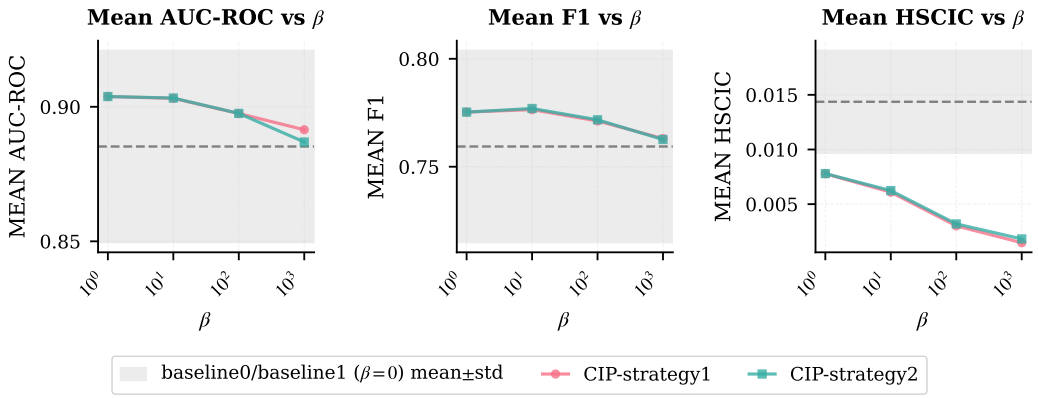


Figure A.97: Public coverage task, fairness CIP features, TX training domain: mean AUC-ROC, Macro-F1 and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

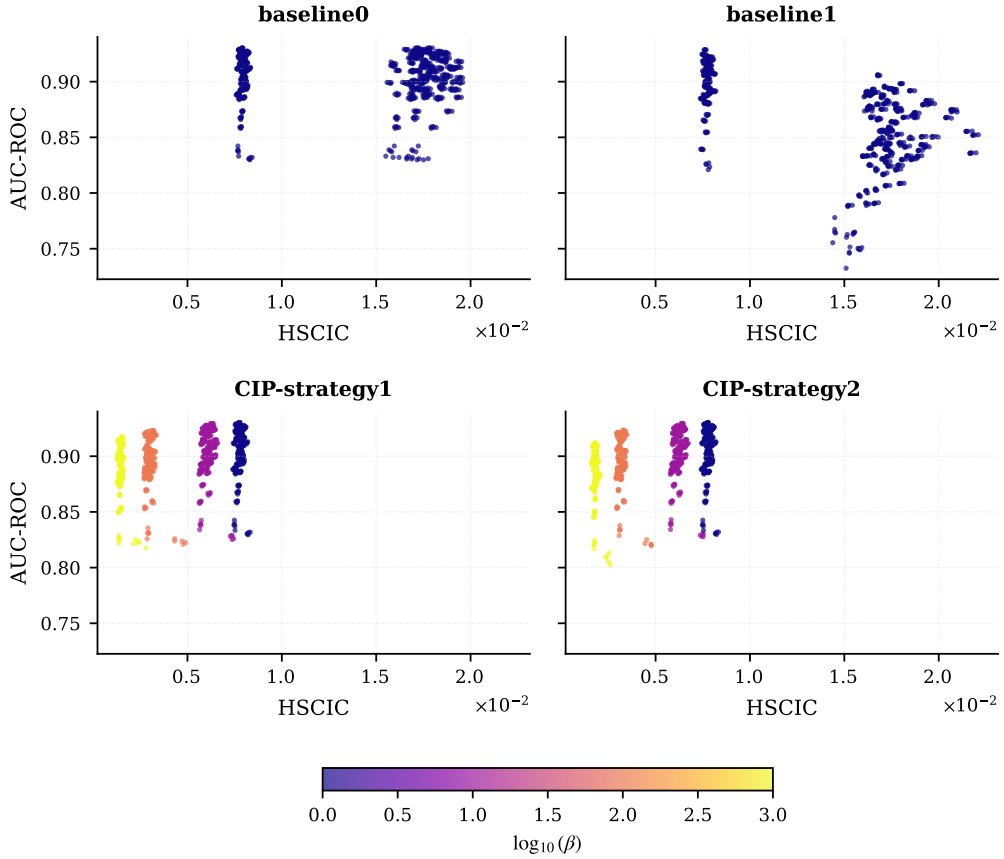


Figure A.98: Public coverage task, fairness CIP features, TX training domain: relation between HSCIC and AUC-ROC across baseline and CIP strategies.

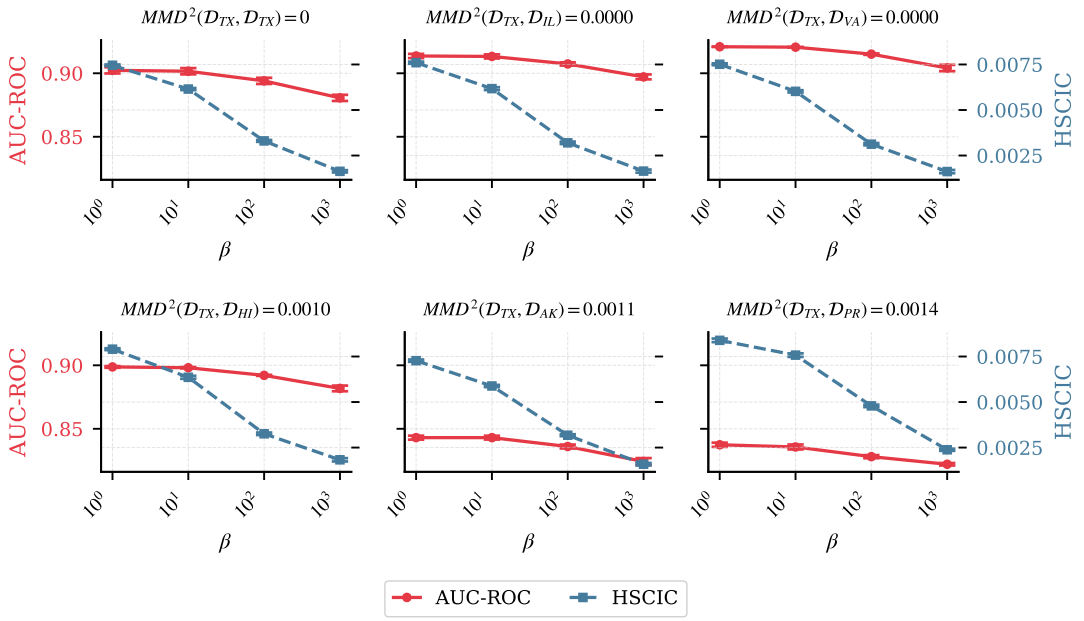


Figure A.99: Public coverage task, fairness CIP features, TX training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

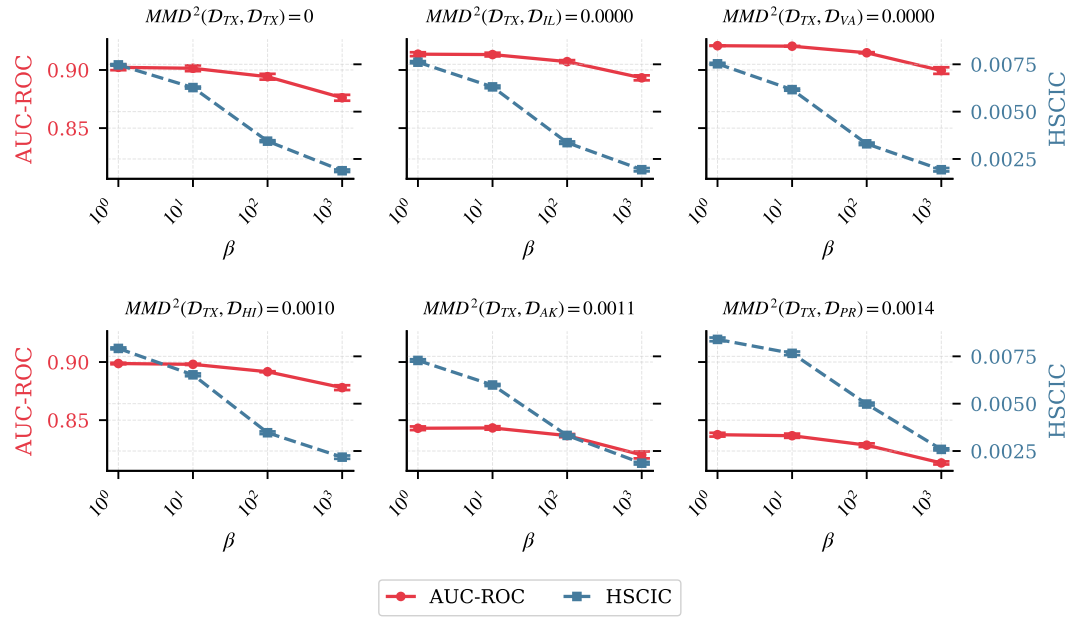


Figure A.100: Public coverage task, fairness CIP features, TX training domain: relation between HSCIC and AUC-ROC versus β for STRATEGY 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

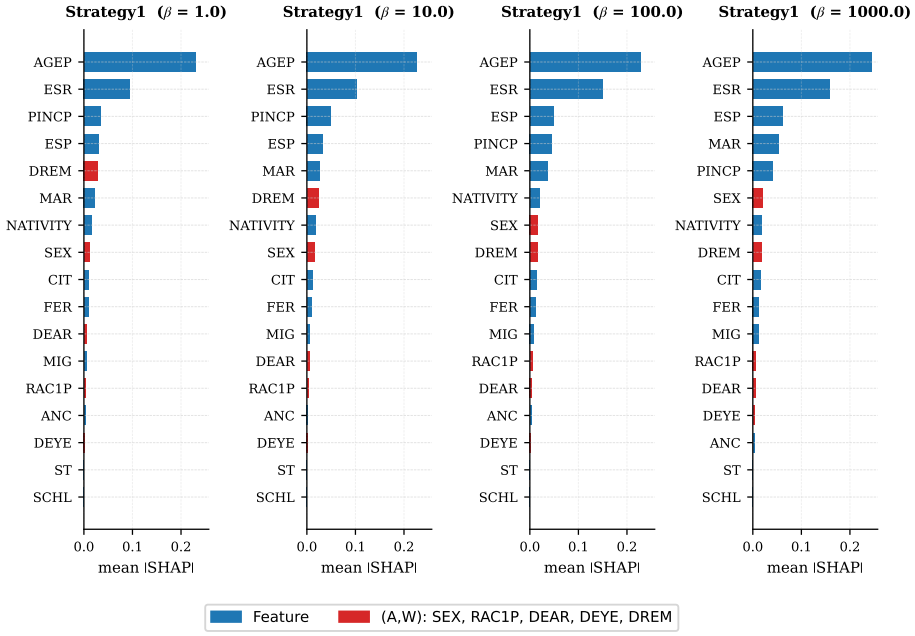


Figure A.101: SHAP analysis: public coverage task, fairness CIP features, TX train domain (out-of-domain), Strategy 1 (joint optimisation). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

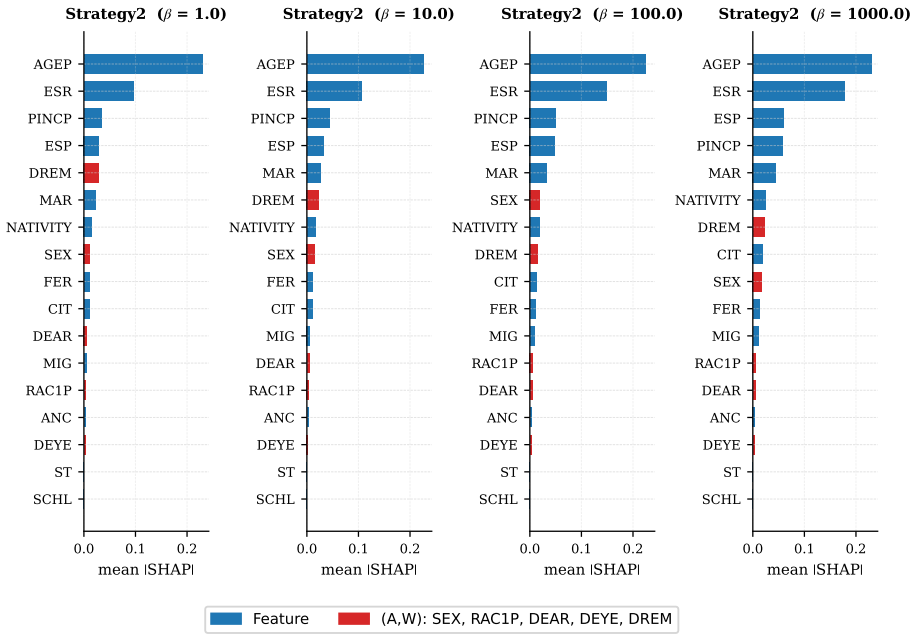


Figure A.102: SHAP analysis: public coverage task, fairness CIP features, TX train domain (out-of-domain), Strategy 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.